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Epilepsy Surgery

Number: 0394

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Scope of Policy

This Clinical Policy Bulletin addresses epilepsy surgery.

I. **Medical Necessity**

Aetna considers the following interventions medically necessary when *all* of the following selection criteria are met:

A. **Epilepsy Surgery**

Cerebral hemispherectomy, corpus callosotomy, and temporal lobectomy (including selective amygdalohippocampectomy) when *all* of the following selection criteria are met:

1. Non-epileptic attacks such as cardiogenic syncope and psychogenic seizures have been ruled out; *and*
2. The diagnosis of epilepsy has been documented, and the epileptic seizure type and syndrome has been clearly defined. In general, appropriate candidates for epilepsy surgery are members who are incapacitated by their frequent seizures as well as the toxicity of anti-epileptic drugs. The general characteristics of individuals for each type of surgical procedure for epilepsy are as follows:

a. **Cerebral hemispherectomy**

Members with unilateral multi-focal epilepsy associated with infantile hemiplegia (especially in hemimegalencephaly and Sturge-Weber disease);

b. **Corpus callosotomy**

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Members with focal to bilateral seizures (formerly known as secondarily generalized seizures);

c. **Temporal lobectomy**

Members with focal impaired awareness seizures (formerly known as complex partial seizures) of temporal or extra-temporal origin; *and*

3. Members' quality of life may significantly improve with surgery; *and*
4. Seizures occur at a frequency that interferes with members' daily living and threatens their well being; *and*
5. The member has tried and failed 2 or more antiepileptic drugs for a minimum of 3 months each absent a contraindication to such a trial. Such a trial must include documentation of compliance.

Aetna considers cerebral hemispherectomy, corpus callosotomy, and temporal lobectomy (including selective amygdalohippocampectomy) experimental, investigational, or unproven when selection criteria are not met.

B. **Deep Brain Stimulation**

Deep brain stimulation for members with intractable seizures when medical necessity criteria are met in CPB 0208 - Deep Brain Stimulation;

C. **Responsive Cortical Stimulation**

Responsive cortical stimulation/responsive neurostimulation (e.g., the NeuroPace RNS System) for adults with intractable focal aware seizures (formerly partial seizures (motor or sensory)) or focal impaired awareness seizures (formerly complex partial seizures) (with motor manifestations)) with or without focal to bilateral seizures (formerly known as secondarily generalized seizures) when the following criteria are met:

1. Non-epileptic attacks such as cardiogenic syncope and psychogenic seizures have been ruled out; *and*
2. The diagnosis of epilepsy has been documented, and the epileptic seizure type and syndrome has been clearly defined. In general, appropriate candidates for responsive cortical stimulation are members who are incapacitated by their frequent seizures as well as the toxicity of anti-epileptic drugs; *and*
3. The member has been diagnosed with no more than two epileptogenic regions; *and*
4. Indications for responsive cortical stimulation - member has *one* of the following indications for responsive cortical stimulation:

    a. Independent onset of left and right temporal lobe onset seizures in persons who are not candidates for resection due to the loss of memory and language that bilateral temporal resection is known to cause; *or*
    b. Left temporal lobe onset seizures where there is concern of language or memory impairment with a resection based upon WADA testing and the rest of the diagnostic work up; *or*
    c. More than one zone of ictal onset, either temporal lobe, neocortical, or both, clearly localized by intracranial recordings, MEG, or other suitable presurgical evaluation making surgical resection unlikely to be successful; *or*
    d. A well-defined neocortical focus for seizures, with or without anatomic abnormality on neuroimaging, either with or without overlap of eloquent cortex; *and*
5. Member has seizures that are severe enough to cause injuries or significantly impair functional ability in domains including employment, psychosocial, education and

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mobility; *and*
6. Members' quality of life may significantly improve with responsive cortical stimulation; *and*
7. There must have been an adequate period of therapy of two or more antiepileptic drugs, namely, the correct drugs used in the correct dosage, carefully monitored for treatment effects and members' compliance; *and*
8. Member does not have an electronic medical device that delivers electrical energy to the head; *and*
9. Member's seizure onset zones are not located below the level of the subthalamic nucleus (lead placement would present too high a risk).

Aetna considers responsive cortical stimulation experimental, investigational, or unproven for primary generalized seizures and for all other indications.

D. **Stereotactic Radiosurgery**

Stereotactic radiosurgery for the treatment of small-volume hypothalamic hamartomas with intractable epilepsy in children with seizures aged over 1 year, hamartoma less than 3 cm^3 and area of fusion with hypothalamus less than 150 mm^2.

Aetna considers the use of stereotactic radiosurgery including radiofrequency amygdalohippocampectomy for medial temporal lobe epilepsy and epilepsy arising in other functional cortical regions experimental, investigational, or unproven because its effectiveness for these indications has not been established.

E. **Magnetic Resonance-guided Laser Interstitial Thermal Therapy**

Magnetic resonance-guided laser interstitial thermal therapy (MRgLITT) (e.g. the NeuroBlate and the Visualase Thermal Therapy System) as an alternative to standard surgery where criteria in section A. on epilepsy surgery are met.

Note: The Wada test (intra-carotid amobarbital procedure), part of the pre-surgical evaluation of members who may undergo temporal lobectomy, is considered a medically necessary service.

II. Experimental, Investigational, or Unproven

Aetna considers the following interventions experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

- Artificial intelligence-assisted technologies (convolutional neural network algorithm, deep learning) for localization of epileptogenic zone;
- Chronic pulvinar stimulation for the treatment of drug-resistant epilepsy;
- Epicranial neurostimulator system for the treatment of epilepsy;
- Examination of genetic variations in members with refractory epilepsy to guide the selection of surgical candidates;
- Hippocampal electrical stimulation for the treatment of mesial temporal lobe epilepsy;
- Intraoperative magnetic resonance imaging in pediatric epilepsy surgery for the treatment of focal cortical dysplasia;
- Localized neocortical resections for uncontrolled focal impaired awareness seizures (formerly complex partial seizures);
- Radiofrequency-thermocoagulation for the treatment of pediatric epilepsy;
- Responsive neurostimulation of the centromedian nucleus of the thalamus for the treatment of pediatric drug-resistant generalized epilepsy;

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- Robotic stereotactic assistance (ROSA) device for the evaluation of the electrode implantation accuracy in epilepsy surgery;
- Stem cell therapy as well as gene therapy for the treatment of refractory epilepsy;
- Subpial transection surgery for refractory epilepsy;
- The use of high-frequency oscillations in epilepsy surgery planning;
- Translocator protein 18 kDa (TSPO) PET imaging in the pre-surgical evaluation of patients with drug-resistant focal epilepsy;
- Trigeminal nerve stimulation for members with intractable seizures.

III. Related Policies

- CPB 0083 - Stereotactic Radiosurgery
- CPB 0191 - Vagus Nerve Stimulation
- CPB 0208 - Deep Brain Stimulation
- CPB 0221 - Quantitative EEG (Brain Mapping)
- CPB 0322 - Electroencephalographic (EEG) Video Monitoring
- CPB 0425 - Ambulatory Electroencephalography
- CPB 0469 - Transcranial Magnetic Stimulation and Cranial Electrical Stimulation
- CPB 0781 - Interstitial Laser Therapy

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
61534	Craniotomy with elevation of bone flap; for excision of epileptogenic focus without electrocorticography during surgery
61536	for excision of epileptic focus, with electrocorticography during surgery
61537	for lobectomy, temporal lobe, without electrocorticography during surgery
61538	for lobectomy with electrocorticography during surgery, temporal lobe
61541	for transection of corpus callosum
61543	for partial or subtotal hemispherectomy
61566	Craniotomy with elevation of bone flap; for selective amygdalohippocampectomy
61736	Laser interstitial thermal therapy (LITT) of lesion, intracranial, including burr hole(s), with magnetic resonance imaging guidance, when performed; single trajectory for 1 simple lesion
61737	multiple trajectories for multiple or complex lesion(s)
61796	Stereotactic radiosurgery (particle beam, gamma ray, or linear accelerator); 1 simple cranial lesion
61797	each additional cranial lesion, simple (List separately in addition to code for primary procedure)
61798	1 complex cranial lesion
61799	each additional cranial lesion, complex (List separately in addition to code for primary procedure)

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Code	Code Description
61800	Application of stereotactic headframe for stereotactic radiosurgery (List separately in addition to code for primary procedure)
61863 - 61864	Twist drill, burr hole, craniotomy, or craniectomy with stereotactic implantation of neurostimulator electrode array in subcortical site (e.g., thalamus, globus pallidus, subthalamic nucleus, periventricular, periaqueductal gray), without use of intraoperative microelectrode recording
61867 - 61868	Twist drill, burr hole, craniotomy, or craniectomy with stereotactic implantation of neurostimulator electrode array in subcortical site (e.g., thalamus, globus pallidus, subthalamic nucleus, periventricular, periaqueductal gray), with use of intraoperative microelectrode recording
61880	Revision or removal of intracranial neurostimulator electrodes[covered for intractable seizures]
61885 - 61886	Insertion or replacement of cranial neurostimulator pulse generator or receiver, direct or inductive coupling
61889	Insertion of skull-mounted cranial neurostimulator pulse generator or receiver, including craniectomy or craniotomy, when performed, with direct or inductive coupling, with connection to depth and/or cortical strip electrode array(s)
61891	Revision or replacement of skull-mounted cranial neurostimulator pulse generator or receiver with connection to depth and/or cortical strip electrode array(s)
64596	Insertion or replacement of percutaneous electrode array, peripheral nerve, with integrated neurostimulator, including imaging guidance, when performed; initial electrode array
64597	each additional electrode array (List separately in addition to code for primary procedure)
77371	Radiation treatment delivery, stereotactic radiosurgery (SRS), complete course of treatment of cranial lesion(s) consisting of 1 session; multi-source Cobalt 60 based
77372	linear accelerator based
77432	Stereotactic radiation treatment management of cranial lesion(s) (complete course of treatment consisting of 1 session)
77435	Stereotactic body radiation therapy, treatment management, per treatment course, to 1 or more lesions, including image guidance, entire course not to exceed 5 fractions
95836	Electrocorticogram from an implanted brain neurostimulator pulse generator/transmitter, including recording, with interpretation and written report, up to 30 days [covered for intractable seizures]
95958	Wada activation test for hemispheric function, including electroencephalographic (EEG) monitoring
95970 - 95971	Electronic analysis of implanted neurostimulator pulse generator system (eg, rate, pulse amplitude, pulse duration, configuration of wave form, battery status, electrode selectability, output modulation, cycling, impedance and patient compliance measurements)
95976 - 95977	Electronic analysis of implanted neurostimulator pulse generator/transmitter (eg, contact group[s], interleaving, amplitude, pulse width, frequency [Hz], on/off cycling, burst, magnet mode, dose lockout, patient selectable parameters, responsive neurostimulation, detection algorithms, closed loop parameters, and passive parameters) by physician or other qualified health care professional
95983	Electronic analysis of implanted neurostimulator pulse generator/transmitter (eg, contact group[s], interleaving, amplitude, pulse width, frequency [Hz], on/off cycling, burst, magnet mode, dose lockout, patient selectable parameters, responsive neurostimulation, detection algorithms, closed loop parameters, and passive parameters) by physician or other qualified health care professional; with brain neurostimulator pulse generator/ transmitter programming, first 15 minutes face-

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Code	Code Description
	hyphento-hyphen face time with physician or other qualified health care professional [covered for intractable seizures]
95984	Electronic analysis of implanted neurostimulator pulse generator/transmitter (eg, contact group[s], interleaving, amplitude, pulse width, frequency [Hz], on/off cycling, burst, magnet mode, dose lockout, patient selectable parameters, responsive neurostimulation, detection algorithms, closed loop parameters, and passive parameters) by physician or other qualified health care professional; with brain neurostimulator pulse generator/ transmitter programming, first 15 minutes face-hyphento-hyphen face time with physician or other qualified health care professional [covered for intractable seizures]

CPT codes not covered for indications listed in the CPB:

***Artificial intelligence (convolutional neural network algorithm, deep learning), Chronic pulvinar stimulation, Translocator protein 18 kDa (TSPO) PET imaging, Intraoperative magnetic resonance imaging* -no specific code**

38232	Bone marrow harvesting for transplantation; autologous
38240	Hematopoietic progenitor cell (HPC); allogeneic transplantation per donor
38241	autologous transplantation
38242	Allogeneic donor lymphocyte infusions
61567	Craniotomy with elevation of bone flap; for multiple subpial transections, with electrocorticography during surgery [subpial transection surgery]
64553	Percutaneous implantation of neurostimulator electrode array; cranial nerve
+95940	Continuous intraoperative neurophysiology monitoring in the operating room, one on one monitoring requiring personal attendance, each 15 minutes (List separately in addition to code for primary procedure)
+95941	Continuous intraoperative neurophysiology monitoring, from outside the operating room (remote or nearby) or for monitoring of more than one case while in the operating room, per hour (List separately in addition to code for primary procedure)

Other CPT codes related to the CPB:

95961 - 95962	Functional cortical and subcortical mapping by stimulation and/or recording of electrodes on brain surface, or of depth electrodes, to provoke seizures or identify vital brain structures

HCPCS codes covered if selection criteria are met:

G0339	Image guided robotic linear accelerator-based stereotactic radiosurgery, complete course of therapy in one session, or first session of fractionated treatment
G0340	Image guided robotic linear accelerator-based stereotactic radiosurgery, delivery including collimator changes and custom plugging, fractionated treatment, all lesions, per session, second through fifth sessions, maximum 5 sessions per course of treatment

HCPCS codes not covered for indications listed in the CPB:

A4541	Monthly supplies for use of device coded at e0733
E0733	Transcutaneous electrical nerve stimulator for electrical stimulation of the trigeminal nerve

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Code	Code Description
G0453	Continuous intraoperative neurophysiology monitoring, from outside the operating room (remote or nearby), per patient, (attention directed exclusively to one patient) each 15 minutes (list in addition to primary procedure)
L8680	Implantable neurostimulator electrode, each
L8681	Patient programmer (external) for use with implantable programmable implantable neurostimulator pulse generator
L8682	Implantable neurostimulator radiofrequency receiver
L8683	Radiofrequency transmitter (external) for use with implantable neurostimulator radiofrequency receiver
L8685	Implantable neurostimulator pulse generator, single array, rechargeable, includes extension
L8688	Implantable neurostimulator pulse generator, dual array, non-rechargeable, includes extension
L8689	External recharging system for battery (internal) for use with implantable neurostimulator
L8695	External recharging system for battery (external) for use with implantable neurostimulator
S2142	Cord blood-derived stem cell transplantation, allogenic
S2150	Bone marrow or blood-derived stem cells (peripheral or umbilical), allogenic or autologous, harvesting, transplantation, and related complications; including; pheresis and cell preparation/storage; marrow ablative therapy; drugs, supplies, hospitalization with outpatient follow-up; medical/surgical, diagnostic, emergency, and rehabilitative services; and the number of days of pre- and post-transplant care in the global definition
S2900	Surgical techniques requiring use of robotic surgical system [Robotic stereotactic assistance (ROSA) device]

Other HCPCS codes related to the CPB:

C9254	Injection, lacosamide, 1 mg
J1165	Injection, phenytoin sodium, per 50 mg
J1953	Injection, phenytoin sodium, per 50 mg
J2060	Injection, lorazepam, 2 mg
J2250	Injection, midazolam hydrochloride, per 1 mg
J2251	Injection, midazolam hydrochloride (wg critical care), not therapeutically equivalent to j2250, per 1 mg
J2252	Injection, midazolam in 0.8% sodium chloride, intravenous, not therapeutically equivalent to j2250, 1 mg
J2253	Injection, midazolam (seizalam), 1 mg
J2560	Injection, phenobarbital sodium, up to 120 mg
J2561	Injection, phenobarbital sodium (sezaby), 1 mg
J3360	Injection, diazepam, up to 5 mg
Q2009	Injection, fosphenytoin, 50 mg phenytoin equivalent
S0078	Injection, fosphenytoin sodium, 750 mg

ICD-10 codes covered if selection criteria are met:

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Code	Code Description
G40.011 - G40.019 G40.111 - G40.119 G40.211 - G40.219 G40.311 - G40.319 G40.A11 - G40.A19 G40.B11 - G40.B19 G40.411 - G40.419 G40.811 - G40.812 G40.911 - G40.919, G40.C11, G40.C19, G40.001 - G40.009 G40.101 - G40.109 G40.201 - G40.209 G40.301 - G40.309 G40.A01 - G40.A09 G40.B01 - G40.B09 G40.401 - G40.409 G40.501 - G40.509 G40.801 - G40.804 G40.901 - G40.909, G40.C01, G40.C09	Epilepsy
Q85.9	Other phakomatoses, not elsewhere classified [small-volume hypothalamic hamartomas]

ICD-10 codes not covered for indications listed in the CPB:

Q04.8	Other specified congenital malformations of brain [Focal cortical dysplasia]

Epicranial neurostimulator:

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Code	Code Description

CPT codes not covered for indications listed in CPB:

0968T	Insertion or replacement of epicranial neurostimulator system, including electrode array and pulse generator, with connection to electrode array
0969T	Removal of epicranial neurostimulator system

ICD-10 codes not covered for indications listed in the CPB:

G40.0-G40.C19	Epilepsy

Radiofrequency-thermocoagulation:

CPT codes not covered for indications listed in CPB:

***Radiofrequency-thermocoagulation*- no specific code**

ICD-10 codes not covered for indications listed in the CPB:

G40.0 - G40.C19	Epilepsy [pediatric]

NeuroPace:

CPT codes covered if criteria are met:

61850	Twist drill or burr hole(s) for implantation of neurostimulator electrodes, cortical
61860	Craniectomy or craniotomy for implantation of neurostimulator electrodes, cerebral, cortical
61863	Twist drill, burr hole, craniotomy, or craniectomy with stereotactic implantation of neurostimulator electrode array in subcortical site (e.g., thalamus, globus pallidus, subthalamic nucleus, periventricular, periaqueductal gray), without use of intraoperative microelectrode recording, first array
61864	each additional array (List separately in addition to primary procedure)
61880	Revision or removal of intracranial neurostimulator electrodes
61885	Insertion or replacement of cranial neurostimulator pulse generator or receiver, direct or inductive coupling; with connection to a single electrode array
61886	with connection to 2 or more electrode arrays
61888	Revision or removal of cranial neurostimulator pulse generator or receiver
95836	Electrocorticogram from an implanted brain neurostimulator pulse generator/transmitter, including recording, with interpretation and written report, up to 30 days
95970	Electronic analysis of implanted neurostimulator pulse generator system (eg, rate, pulse amplitude, pulse duration, configuration of wave form, battery status, electrode selectability, output modulation, cycling, impedance and patient compliance measurements); simple or complex brain, spinal cord, or peripheral (ie, cranial nerve, peripheral nerve, sacral nerve, neuromuscular) neurostimulator pulse generator/transmitter, without programming

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Code	Code Description
95971	simple spinal cord, or peripheral (ie, peripheral nerve, sacral nerve, neuromuscular) neurostimulator pulse generator/transmitter, with intraoperative or subsequent programming
95977	Electronic analysis of implanted neurostimulator pulse generator/transmitter (eg, contact group[s], interleaving, amplitude, pulse width, frequency [Hz], on/off cycling, burst, magnet mode, dose lockout, patient selectable parameters, responsive neurostimulation, detection algorithms, closed loop parameters, and passive parameters) by physician or other qualified health care professional; with complex cranial nerve neurostimulator pulse generator/transmitter programming by physician or other qualified health care professional
95983	Electronic analysis of implanted neurostimulator pulse generator/transmitter (eg, contact group[s], interleaving, amplitude, pulse width, frequency [Hz], on/off cycling, burst, magnet mode, dose lockout, patient selectable parameters, responsive neurostimulation, detection algorithms, closed loop parameters, and passive parameters) by physician or other qualified health care professional; with brain neurostimulator pulse generator/ transmitter programming, first 15 minutes face-to-face time with physician or other qualified health care professional
95984	Electronic analysis of implanted neurostimulator pulse generator/transmitter (eg, contact group[s], interleaving, amplitude, pulse width, frequency [Hz], on/off cycling, burst, magnet mode, dose lockout, patient selectable parameters, responsive neurostimulation, detection algorithms, closed loop parameters, and passive parameters) by physician or other qualified health care professional; with brain neurostimulator pulse generator/ transmitter programming, each additional 15 minutes face-to-face time with physician or other qualified health care professional (List separately in addition to code for primary procedure)

CPT codes not covered for indications listed in CPB:

***Responsive neurostimulation of the centromedian nucleus of the thalamus*- no specific code**

HCPCS codes covered if criteria are met :

C1767	Generator, neurostimulator (implantable), non-rechargeable
L8687	Implantable neurostimulator pulse generator, dual array, rechargeable, includes extension
L8688	Implantable neurostimulator pulse generator, dual array, non-rechargeable, includes extension

ICD-10 codes covered if selection criteria are met:

G40.011 - G40.019	Localization-related (focal) (partial) idiopathic epilepsy and epileptic syndromes with seizures of localized onset, intractable
G40.111 - G40.119	Localization-related (focal) (partial) symptomatic epilepsy and epileptic syndromes with simple partial seizures, intractable
G40.211 - G40.219	Localization-related (focal) (partial) symptomatic epilepsy and epileptic syndromes with complex partial seizures, intractable

ICD-10 codes not covered for indications listed in the CPB:

G40.311-G40.319	Generalized idiopathic epilepsy and epileptic syndromes, intractable
G40.411-G40.419	Other generalized epilepsy and epileptic syndromes, intractable

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Code	Code Description
G40.A11-G40.A19	Absence epileptic syndrome, intractable
G40.B11-G40.B19	Juvenile myoclonic epilepsy, intractable
G40.C11-G40.C19	Lafora progressive myoclonus epilepsy, intractable

***Magnetic resonance-guided laser interstitial thermal therapy (e.g. the NeuroBlate and the Visualase Thermal Therapy System* - no specific code :**

Background

For patients who have intractable seizures despite adequate treatment with appropriate antiepileptic drugs, surgery is their last hope. The goal of epilepsy surgery is not only to decrease the frequency of seizures, but also to improve quality of life.

Temporal lobectomy has been found to be safe and effective for treating patients with complex partial seizures of temporal or extratemporal origin. Patients who have a single identifiable focus in a restricted cortical area that can be safely excised without producing additional disability can be considered as candidates for temporal lobectomy.

Corpus callosotomy has been found to be safe and effective for treating patients with partial and secondarily generalized seizures.

There is only limited evidence that cerebral hemispherectomy is effective in managing unilateral multi-focal epilepsy associated with infantile hemiplegia (especially in hemimegalencephaly and Sturge-Weber disease). However, it is the last hope for these patients to eliminate/alleviate their disabling epileptic seizures, and to avoid adverse irreversible psychosocial consequences that may lead to lifelong disability.

Since the advent of deep brain stimulation (DBS) for the treatment of a variety of movement disorders, studies have been performed to ascertain whether this method can reduce seizure frequency. Evidence from experimental animal studies suggests the existence of a nigral control of the epilepsy system. The results of animal studies are promising, but work on humans is preliminary.

In a pilot study, Boon et al (2007) assessed the effectiveness of long-term DBS in medial temporal lobe (MTL) structures in patients with MTL epilepsy. A total of 12 consecutive patients with refractory MTL epilepsy were included in this study. The protocol included invasive video-EEG monitoring for ictal-onset localization and evaluation for subsequent stimulation of the ictal-onset zone. Side effects and changes in seizure frequency were carefully monitored. Ten of 12 patients underwent long-term MTL DBS; 2 of 12 patients underwent selective amygdalo-hippocampectomy. After mean follow-up of 31 months (range of 12 to 52 months), 1 of 10 stimulated patients was seizure-free (more than 1 year), 1 of 10 patients had a greater than 90 % reduction in seizure frequency; 5 of 10 patients had a seizure-frequency reduction of greater than equal to 50 %; 2 of 10 patients had a seizure-frequency reduction of 30 to 49 %; and 1 of 10 patients was a non-responder. None of the patients reported side effects. In 1 patient, magnetic resonance imaging (MRI) showed asymptomatic intra-cranial hemorrhages along the trajectory of the DBS electrodes. None of the patients showed changes in clinical neurological testing. Patients who underwent selective amygdalo-hippocampectomy are seizure-free (more than 1 year), anti-epileptic drugs are unchanged, and no side effects have occurred. The authors concluded that this open pilot study demonstrated the potential efficacy of long-term DBS in MTL structures that should now be further confirmed by multi-center randomized controlled trials (RCTs).

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The Wada test (intra-carotid amytal procedure) is commonly used as a predictor of memory dysfunction following temporal lobectomy for intractable epilepsy. Asymmetry in memory scores can provide focus lateralizing information.

The Agency for Healthcare Research and Quality's technology assessment on the management of treatment-resistant epilepsy stated that the data are inconsistent across studies and do not allow for firm evidence-based conclusions as to the exact proportion of patients who will become seizure-free or who will not benefit from multiple subpial transection. In addition, too few studies were available to allow for an evidence-based evaluation of parietal or occipital lobe surgery (Chapell et al, 2003). The American Academy of Neurology (AAN)'s practice parameter on temporal lobe and localized neocortical resections for epilepsy stated that there remains no Class I or II evidence regarding the safety and efficacy of localized neocortical resections. Further studies are needed to determine if neocortical seizures benefit from surgery (Engel et al, 2003).

Candidates for epilepsy surgery and their family, if applicable, should receive detailed information regarding the specific surgical procedures and their possible benefits and side effects. Candidates for epilepsy surgery should not have co-existent progressive neurological disease or major psychological or medical disorder. Persons with progressive neurological diseases or major medical or psychological disorders are generally unsuitable candidates for epilepsy surgery because of the possibility that surgery could worsen the course of these other conditions.

In a pilot study (n = 5), Velasco and colleagues (2005) examined the safety and effectiveness of cerebellar stimulation (CS) on patients with medically refractory motor seizures, and especially generalized tonic-clonic seizures. Bilateral modified 4-contact plate electrodes were placed on the cerebellar superomedial surface through 2 sub-occipital burr holes. The implanted programmable, battery-operated stimulator was adjusted to 2.0 microC/cm2/phase with the stimulator case as the anode; at this level, no patient experienced the stimulation. Patients served as their own controls, comparing their seizure frequency in pre-implant basal phase (BL) of 3 months with the post-implant phases from 10 months to 4 years (average, 8 epochs of 3 months each). During the month after implantation, the stimulators were not activated. The patient and the evaluator were blinded as to the next 3-month epoch, as to whether stimulation was used. The patients were randomized into 2 groups:

I. 3 with the stimulator ON, and
II. 2 with the stimulator OFF.

I. 3 with the stimulator ON, and
II. 2 with the stimulator OFF.

After a 4-month post-implantation period, all patients had their stimulator ON until the end of the study and beyond. Medication was maintained unchanged throughout the study. EEG paroxysmal discharges also were measured. Generalized tonic-clonic seizures: in the initial 3-month double-blind phase, 2 patients were monitored with the stimulation OFF; no change was found in the mean seizure rate (patient 1, 100 %, and patient 5, 85 %; mean, 93 %), whereas the 3 patients with the stimulation initially ON had a reduction of seizures to 33 % (patient 2, 21 %; patient 3, 46 %; patient 4, 32 %) with a statistically significant difference between OFF and ON phase of $p = 0.023$. All 5 patients then were stimulated and monitored. At the end of the next 6 months of stimulation, the 5 patients had a mean seizure rate of 41 % (14 to 75 %) of the BL. The second patient developed an infection in the implanted system, which had to be removed after 11 months of stimulation; the seizures were being reduced with stimulation to a mean of 1 per month from a mean of 4.7 per month (BL level) before stimulation. At the end of 24 months, 3 patients were monitored with stimulation, resulting in a further reduction of seizures to 24 % (11 to 38 %). Tonic seizures: 4 patients had these seizures, which at 24 months were reduced to 43 % (10 to 76 %). Follow-up surgery was necessary in 4 patients because of infection in 1 patient and lead/electrode displacement needing repositioning in 3 patients. The statistical analysis showed a significant reduction in tonic-clonic seizures ($p < 0.001$) and tonic seizures ($p < 0.05$). These investigators concluded that the superomedial cerebellar cortex appears to be a safe and effective target for electrical stimulation for decreasing motor seizures over the long-term. The effect shows generalized tonic-clonic seizure reduction after 1 to 2 months and continues to decrease over the

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first 6 months and then maintains this effectiveness over the study period of 2 years and beyond. The results of this pilot study needed to be validated by additional trials with larger patient populations.

Fountas et al (2010) reviewed the pertinent literature to outline the role of CS in the management of medically refractory epilepsy. The pertinent articles were categorized into 2 large groups:

I. animal experimental and
II. human clinical studies.

I. animal experimental and
II. human clinical studies.

Particular emphasis on the following aspects was given when reviewing the human clinical studies: their methodological characteristics, the number of participants, their seizure types, the implantation technique and its associated complications, the exact stimulation target, the stimulation technique, the seizure outcome, and the patients' psychological and social post-stimulation status. Three clinical double-blind studies were found, with similar implantation surgical technique, stimulation target, and stimulation parameters, but quite contradictory results. Two of these studies failed to demonstrate any significant seizure reduction, whereas the third one showed a significant post-stimulation decrease in seizure frequency. All possible factors responsible for these differences in the findings were analyzed in the present study. The authors concluded that CS seems to remain a stimulation target worth exploring for defining its potential in the treatment of medically intractable epilepsy, although the data from the double-blind clinical studies that were performed failed to establish a clear benefit in regard to seizure frequency. They noted that a large-scale, double-blind clinical study is needed for accurately defining the efficacy of CS in epilepsy treatment.

Electrical stimulation of the hippocampus has been proposed as a possible treatment for mesial temporal lobe epilepsy (MTLE). Tellez-Zenteno et al (2006) reported their findings of 4 patients with refractory MTLE (whose risk to memory contraindicated temporal lobe resection) who underwent implantation of a chronic stimulating depth electrode along the axis of the left hippocampus. These investigators used continuous, sub-threshold electrical stimulation (90 microsec, 190 Hz) and a double-blind, multiple cross-over, randomized controlled design, consisting of 3 treatment pairs, each containing two 1-month treatment periods. During each treatment pair, the stimulator was randomly turned ON 1 month and OFF 1 month. Outcomes were assessed at monthly intervals in a double-blind manner, using standardized instruments and accounting for a washout period. These researchers compared outcomes between ON, OFF, and baseline periods. Hippocampal stimulation produced a median reduction in seizures of 15 %. All but 1 patient's seizures improved; however, the results did not reach significance. Effects seemed to carry over into the OFF period, and an implantation effect can not be ruled out. These researchers found no significant differences in other outcomes. There were no adverse effects. One patient has been treated for 4 years and continued to experience substantial long-term seizure improvement. The authors demonstrated important beneficial trends, some long-term benefits, and absence of adverse effects of hippocampal electrical stimulation in MTLE. However, the effect sizes observed were smaller than those reported in non-randomized, unblinded studies. They stated that large scale, double-blind RCTs are needed to ascertain the effectiveness of hippocampal electrical stimulation in patients with MTLE.

Velasco and colleagues (2007) evaluated the safety and effectiveness of electrical stimulation of the hippocampus in a long-term follow-up study, as well as its impact on memory performance in the treatment of patients with refractory MTLE. A total of 9 patients were included. All had refractory partial complex seizures, some with secondary generalizations. All patients had a 3-month-baseline-seizure count, after which they underwent bilateral hippocampal diagnostic electrode implantation to establish focus laterality and location – 3 patients had bilateral; 6 had unilateral foci. Diagnostic electrodes were explanted and definitive Medtronic electrodes were implanted directed into the hippocampal foci. Position was confirmed with MRI and afterwards, the DBS system internalized. Patients attended a medical appointment every 3 months for seizure diary collection, DBS system checkup, and neuropsychological testing. Follow-up ranged from 18 months to 7 years. Patients were divided in 2 groups:

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I. 5 had normal MRIs and seizure reduction of greater than 95 %, and
II. 4 had hippocampal sclerosis and seizure reduction of 50 to 70 %.

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No patient had neuropsychological deterioration, nor did any patient show side effects. Three patients were explanted after 2 years due to skin erosion in the trajectory of the system. The authors concluded that electrical stimulation of the hippocampus provides a non-lesional method that improves seizure outcome without memory deterioration in patients with hippocampal epileptic foci. This is a small study; its findings need to be validated by studies with larger patient populations.

Sun and associates (2008) stated that with the success of DBS for treatment of movement disorders, brain stimulation has received renewed attention as a potential treatment option for epilepsy. Responsive stimulation aims to suppress epileptiform activity by delivering stimulation directly in response to electrographic activity. Animal and human data support the concept that responsive stimulation can abort epileptiform activity, and this modality may be a safe and effective treatment option for epilepsy. Responsive stimulation has the advantage of specificity. In contrast to the typically systemic administration of pharmacotherapy, with the concomitant possibility of side effects, electrical stimulation can be targeted to the specific brain regions involved in the seizure. In addition, responsive stimulation provides temporal specificity. Treatment is provided as needed, potentially reducing the likelihood of functional disruption or habituation due to continuous treatment. The authors reviewed current animal and human research in responsive brain stimulation for epilepsy and discussed the NeuroPace RNS System, an investigational implantable responsive neurostimulator system that is being evaluated in a multi-center, randomized, double-blinded trial to assess the safety and efficacy of responsive stimulation for the treatment of medically refractory epilepsy.

Morrell et al (2011) evaluated the safety and effectiveness of responsive cortical stimulation as an adjunctive therapy for partial onset seizures in adults with medically refractory epilepsy. A total of 191 adults with medically intractable partial epilepsy were implanted with a responsive neurostimulator connected to depth or subdural leads placed at 1 or 2 pre-determined seizure foci. The neurostimulator was programmed to detect abnormal electrocorticographic activity. One month after implantation, subjects were randomized 1:1 to receive stimulation in response to detections (treatment) or to receive no stimulation (sham). Safety and effectiveness were assessed over a 12-week blinded period and a subsequent 84-week open-label period during which all subjects received responsive stimulation. Seizures were significantly reduced in the treatment (-37.9 %, n = 97) compared to the sham group (-17.3 %, n = 94; $p = 0.012$) during the blinded period and there was no difference between the treatment and sham groups in adverse events. During the open-label period, the seizure reduction was sustained in the treatment group and seizures were significantly reduced in the sham group when stimulation began. There were significant improvements in overall quality of life ($p < 0.02$) and no deterioration in mood or neuropsychological function. The authors concluded that responsive cortical stimulation reduces the frequency of disabling partial seizures, is associated with improvements in quality of life, and is well-tolerated with no mood or cognitive effects. They noted that responsive stimulation may provide another adjunctive treatment option for adults with medically intractable partial seizures. However, with its more invasive surgical component, this approach (responsive cortical stimulation) carries greater risks and requires careful patient selection; identification of factors prdicting good outcome prior to electrode implantation would be of great value. Furthermore, responsive cortical stimulation has yet to be approved for use in the U.S.

Gamma knife (GK) radiosurgery has been proposed as an alternative to classic microsurgery in MTLE. Bartolomei and colleagues (2008) reported the efficacy and tolerance of GK radiosurgery in MTLE after a follow-up of more than 5 years. A total of 15 patients were included in this study; 8 were treated on the left side, and 7 were treated on the right. The mean follow-up was 8 years (range of 6 to 10 years). At the last follow-up, 9 of 16 patients (60 %) were considered seizure-free (Engel Class I) (4/16 in Class IA, 5/16 in Class IB). Seizure cessation occurred with a mean delay of 12 months (+/- 3) after GK radiosurgery, often preceded by a period of increasing aura or seizure occurrence (6/15 patients). The mean delay of appearance of the first neuroradiological changes was 12 months (+/- 4). Nine patients (60 %) experienced mild

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headache and were placed on corticosteroid treatment for a short period. All patients who were initially seizure-free experienced a relapse of isolated aura (10/15, 66 %) or complex partial seizures (10/15, 66 %) during anti-epileptic drug tapering. Restoration of treatment resulted in good control of seizures.

In an editorial that accompanied the afore-mentioned paper, Spencer (2008) stated that "gamma knife treatment in mesial temporal lobe epilepsy, then, is still searching for a place. Right now, its disadvantages (slightly lower seizure response rate, delayed response, absolute requirement for continued medications, higher mortality) compared to anterior medial temporal resection seem to outweigh its noninvasive status, which so far does not appear to carry any clear benefits in terms of neurologic or cognitive function, or seizure response. Whether gamma knife treatment should be considered for intractable epilepsy arising in other functional cortical regions that can not be treated with resection remains unexplored. Its efficacy, as well as morbidity, in those situations has not been examined, and the volume and definition of the tissues to be targeted are considerably less well-defined than for mesial lobe epilepsy".

In a pilot study, Barbaro et al (2009) reported the 3-year outcomes of a multi-center, study of GK radiosurgery for MTLE. Radiosurgery was randomized to 20 or 24 Gy targeting the amygdala, hippocampus, and parahippocampal gyrus. Seizure diaries evaluated the final seizure remission between months 24 and 36. Verbal memory was evaluated at baseline and 24 months with the Wechsler Memory Scale-Revised (WMS-R) and California Verbal Learning Test (CVLT). Patients were classified as having "significant improvement," "no change," and "significant impairment" based on relative change indices. Thirteen high-dose and 17 low-dose patients were treated. Both groups showed significant reductions in seizures by 1 year after treatment. At the 36-month follow-up evaluation, 67 % of patients were seizure-free for the prior 12 months (high-dose: 10/13, 76.9 %; low-dose: 10/17, 58.8 %). Use of steroids, headaches, and visual field defects did not differ by dose or seizure remission. The prevalence of verbal memory impairment was 15 % (4/26 patients); none declined on more than 1 measure. The prevalence of significant verbal memory improvements was 12 % (3/26). The authors concluded that GK radiosurgery for unilateral MTLE offers seizure remission rates comparable with those reported previously for open surgery. There were no major safety concerns with high-dose radiosurgery compared with low-dose radiosurgery. They stated that additional research is needed to determine if GK radiosurgery may be a treatment option for some patients with MTLE.

Vojtech et al (2009) examined the effectiveness of GK radiosurgery in the treatment of MTLE due to mesial temporal sclerosis. A total of 14 patients underwent radiosurgical entorhino-amygdalo-hippocampectomy with a marginal dose of 18-, 20-, or 25-Gy to the 50 % isodose following a standard pre-operative epilepsy evaluation. One patient was classified as Engel Class Ib, 3 were Engel Class IIc, 1 was Engel Class IIIa, and 2 were Engel Class IVb in a subgroup of 7 patients who were unoperated 2 years prior to the last visit and at least 8 years after irradiation (average of 116 months). The insufficient effect of irradiation led these investigators to perform epilepsy surgery on another 7 patients an average of 63.5 months after radiosurgery. The average follow-up period was 43.5 months after the operation. Four patients are seizure-free; 1 is Engel Class IIb and 1 is Engel Class IId. One patient can not be classified due to the short period of follow-up. The frequency of seizures tended to rise after irradiation in some patients. Collateral edema was observed in 9 patients, which started earlier and was more frequent in those irradiated with higher doses. It had a marked expansive character in 3 cases and clinical signs of intra-cranial hypertension were present in 3 cases. Partial upper lateral quadrant anopia as a permanent side effect was observed in 2 patients. Repeated psychotic episodes (2 patients) and status epilepticus (2 patients) were also seen after treatment. No significant memory changes occurred in the group as a whole. The authors concluded that radiosurgery with 25-, 20, or 18-Gy marginal dose levels did not lead to seizure control in this patient series, although subsequent epilepsy surgery could stop seizures. Higher doses were associated with the risk of brain edema, intra-cranial hypertension, and a temporary increase in seizure frequency.

Malikova et al (2009) described MRI changes following stereotactic radiofrequency amygdalohippocampectomy (AHE) and correlated the hippocampal and amygdalar volumes reduction with the clinical seizure outcome. A total of 18 patients were included. Volumetry was calculated from pre-operative MRI and from MRI obtained 1 year after the operation. The clinical outcome was examined 1 and 2 years after the treatment. Hippocampal volume decreased by 54 +/- 19 %, and amygdalar volume decreased by 49 +/- 18 %. One year after the procedure, 13 (72 %) patients were classified as Engel's Class I

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(9 as Class IA), 4 (22 %) patients as Class II and 1 (6 %) patient as Class III. Two years after the operation, 14 patients (82 %) were classified as Class I (7 as Class IA) and 3 patients (18 %) as Class II. There were 3 surgical complications after the procedure: 1 small subdural hematoma, and twice a small electrode tip left in operation field (these patients were excluded from the study). In 3 patients, temporary meningeal syndrome developed. The authors concluded that results of stereotactic radiofrequency AHE are promising.

Tellez-Zenteno and Wiebe (2011) stated that hippocampal stimulation should be regarded as an experimental therapy for epilepsy, and patients considered for this intervention should do so in the context of a well-designed RCT. The authors concluded that only well-conducted, blinded, randomized trials, followed by long-term systematic observation will yield a clear picture of the effect of this promising therapy, and will help guide its future use.

In a pilot feasibility study, Degiorgio et al (2006) evaluated the safety and preliminary effectiveness of trigeminal nerve stimulation (TNS) of the infra-orbital and supra-orbital branches of the trigeminal nerve for the treatment of epilepsy. Trigeminal nerve stimulation was well-tolerated. Four (57 %) of 7 subjects who completed greater than or equal to 3 months experienced a greater than or equal to 50 % reduction in seizure frequency. The authors concluded that the results of this pilot study supported further investigation into the safety and effectiveness of TNS for epilepsy.

In a double-blind, randomized controlled trial, Degiorgio et al (2013) examined the safety and effectiveness of external TNS (eTNS) in patients with drug-resistant epilepsy (DRE), and tested the suitability of treatment and control parameters in preparation for a phase III multi-center clinical trial. A total of 50 subjects with 2 or more partial onset seizures per month (complex partial or tonic-clonic) entered a 6-week baseline period, and then were evaluated at 6, 12, and 18 weeks during the acute treatment period. Subjects were randomized to treatment (eTNS 120 Hz) or control (eTNS 2 Hz) parameters. At entry, subjects were highly drug-resistant, averaging 8.7 seizures per month (treatment group) and 4.8 seizures per month (active controls). On average, subjects failed 3.35 anti-epileptic drugs prior to enrollment, with an average duration of epilepsy of 21.5 years (treatment group) and 23.7 years (active control group), respectively. External TNS was well-tolerated. Side effects included anxiety (4 %), headache (4 %), and skin irritation (14 %). The responder rate, defined as greater than 50 % reduction in seizure frequency, was 30.2 % for the treatment group versus 21.1 % for the active control group for the 18-week treatment period (not significant, $p = 0.31$, generalized estimating equation [GEE] model). The treatment group experienced a significant within-group improvement in responder rate over the 18-week treatment period (from 17.8 % at 6 weeks to 40.5 % at 18 weeks, $p = 0.01$, GEE). Subjects in the treatment group were more likely to respond than patients randomized to control (odds ratio 1.73, confidence interval [CI]: 0.59 to 0.51). External TNS was associated with reductions in seizure frequency as measured by the response ratio ($p = 0.04$, analysis of variance [ANOVA]), and improvements in mood on the Beck Depression Inventory ($p = 0.02$, ANOVA). The authors concluded that the findings of this study provided preliminary evidence that eTNS is safe and may be effective in subjects with DRE. Side effects were primarily limited to anxiety, headache, and skin irritation. They stated that these results will serve as a basis to inform and power a larger multi-center phase III clinical trial.

In an editorial that accompanied the afore-mentioned study by Degiorgio et al, Faught and Tatum (2013) stated that “The beneficial effect demonstrated by Degiorgio et al was modest, but is sufficient to encourage design of a more definitive study”.

Liu and associates (2013) stated that with an annual incidence of 50/100,000 people, nearly 1 % of the population suffers from epilepsy. Treatment with anti-epileptic medication fails to achieve seizure remission in 20 to 30 % of patients. One treatment option for refractory epilepsy patients who would not otherwise be surgical candidates is electrical stimulation of the brain, which is a rapidly evolving and reversible adjunctive therapy. Therapeutic stimulation can involve direct stimulation of the brain nuclei or indirect stimulation of peripheral nerves. There are 3 stimulation modalities that have class I evidence supporting their uses:

I. vagus nerve stimulation (VNS),
II. stimulation of the anterior nuclei of the thalamus (ANT), and,
III. the most recently developed, responsive neurostimulation (RNS).

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II. stimulation of the anterior nuclei of the thalamus (ANT), and,
III. the most recently developed, responsive neurostimulation (RNS).

While the other treatment modalities outlined deliver stimulation regardless of neuronal activity, the RNS administers stimulation only if triggered by seizure activity. The lower doses of stimulation provided by such responsive devices can not only reduce power consumption, but also prevent adverse reactions caused by continuous stimulation, which include the possibility of habituation to long-term stimulation. Responsive neurostimulation, as an investigational treatment for medically refractory epilepsy, is currently under review by the Food and Drug Administration.

Ge and colleagues (2013) reviewed the targets of the deep brain and RNS to identify the best optimal stimulation parameters and the best mode of stimulation, whether cyclical, continuous, or smarter. This review was based on data obtained from published articles from 1950 to 2013. To perform the PubMed literature search, the following keywords were input: deep brain stimulation (DBS), RNS, and refractory epilepsy. Articles containing information related to brain stimulation or RNS for the treatment of refractory epilepsy were selected. The currently available treatment options for those patients who resist multiple anti-epileptic medications and surgical procedures include electric stimulation, both direct and indirect, of brain nuclei thought to be involved in epileptogenesis. The number of potential targets has increased over the years to include the ANT, the centromedian nucleus of the thalamus, the hippocampus, the subthalamic nucleus, the caudate nucleus, and the cerebellum, among others. The results of a RCT and the RNS trial were published to reveal the effectiveness. The authors concluded that although statistically significant reductions in seizures had been observed using several different stimulation techniques, including VNS, DBS, and RNS, these effects are currently only palliative and do not approach the effectiveness comparable with that seen in resection in appropriately selected patients. They stated that more research is needed to determine optimal stimulation targets and techniques as well as to determine which epilepsy patients will benefit most from this technology.

Krishnaiah and co-workers (2013) stated that nearly 30 % of patients with epilepsy continue to have seizures in spite of several anti-epileptic drug (AED) regimens. In such cases they are regarded as having refractory, or uncontrolled epilepsy. There is no universally accepted definition for uncontrolled or medically refractory epilepsy, but for the purpose of this review, these investigators considered seizures to be drug resistant if they failed to respond to a minimum of 2 AEDs. It is believed that early surgical intervention may prevent seizures at a younger age and improve the intellectual and social status of children. There are many types of surgery for refractory epilepsy with subpial transection being one. In a Cochrane review, these researchers determined the benefits and adverse effects of subpial transection for partial-onset seizures and generalized tonic-clonic seizures in children and adults. They searched the Cochrane Epilepsy Group Specialised Register (August 8, 2013), the Cochrane Central Register of Controlled Trials (CENTRAL Issue 7 of 12, The Cochrane Library July 2013), and MEDLINE (1946 to August 8, 2013). They did not impose any language restrictions. These investigators considered all randomized and quasi-randomized parallel group studies either blinded or non-blinded. Two review authors independently screened the trials identified by the search. The same 2 authors planned to independently assess the methodological quality of studies. If studies had been identified for inclusion, 1 author would have extracted the data and the other would have verified it. No relevant studies were found. The authors concluded that there is no evidence to support or refute the use of subpial transection surgery for medically refractory cases of epilepsy. Moreover, they stated that well-designed RCTs are needed to guide clinical practice.

Gloss and colleagues (2014) stated that approximately 2/3 of seizures can be controlled with anti-epileptic medications. For some of the others, surgery can completely eliminate or significantly reduce the occurrence of disabling seizures. Localization of epileptogenic areas for resective surgery is far from perfect, and new tools are being investigated to more accurately localize the epileptogenic zone and improve the likelihood of freedom from post-surgical seizures. Recordings of pathological high-frequency oscillations (HFOs) may be one such tool. In a Cochrane review, these investigators evaluated the ability of HFOs to improve the outcomes of epilepsy surgery by helping to identify more accurately the epileptogenic areas of the brain. They searched the Cochrane Epilepsy Group Specialized Register (April 15, 2013), the Cochrane Central Register of Controlled Trials (CENTRAL) in The Cochrane Library (2013, Issue 3), MEDLINE (Ovid) (1946

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to April 15, 2013), CINAHL (EBSCOhost) (April 15, 2013), Web of Knowledge (Thomson Reuters) (April 15, 2013), www.clinicaltrials.gov (April 15, 2013), and the World Health Organization International Clinical Trials Registry Platform (April 15, 2013). These researchers included studies that provided information on the outcomes of epilepsy surgery at 6 months or more and which used HFOs in making decisions about epilepsy surgery. The primary outcome of the review was the Engel Class Outcome System. Secondary outcomes were responder rate, International League Against Epilepsy (ILAE) epilepsy surgery outcome, frequency of adverse events from any source and quality of life outcomes. They intended to analyze outcomes via an aggregated data fixed-effect model meta-analysis. Two studies met the inclusion criteria. Both studies were small non-randomized trials, with no control group and no blinding. The quality of evidence for all outcomes was very low. The combination of these 2 studies resulted in 11 participants who prospectively used ictal HFOs for epilepsy surgery decision making. Results of the post-surgical seizure freedom Engel class I to IV outcome were determined over a period of 12 to 38 months (average of 23.4 months) and indicated that 6 participants had an Engel class I outcome (seizure freedom), 2 had class II (rare disabling seizures), 3 had class III (worthwhile improvement). No adverse effects were reported. Neither study compared surgical results guided by HFOs versus surgical results guided without HFOs. The authors concluded that no reliable conclusions can be drawn regarding the effectiveness of using HFOs in epilepsy surgery decision making at present.

The NeuroPace RNS System is a responsive cortical stimulator for the treatment of medically intractable partial epilepsy. The RNS System includes a cranially implanted programmable neurostimulator that is connected to one or two depth and/or subdural cortical strip leads that are surgically placed in or on the brain at the seizure focus. The neurostimulator continuously senses brain electrical activity through the leads. When abnormal brain electrical activity typical of the activity that precedes that patient's seizures is detected, the neurostimulator delivers pulses of stimulation through those same electrodes before an individual experiences seizures.

Heck et al (2014) sought to evaluate the safety and effectiveness of responsive stimulation at the seizure focus as an adjunctive therapy to reduce the frequency of seizures in adults with medically intractable partial onset seizures arising from 1 or 2 seizure foci. The investigators conducted a randomized multi-center double-blinded controlled trial of responsive focal cortical stimulation (RNS System). Subjects with medically intractable partial onset seizures from 1 or 2 foci were implanted, and 1 month post-implant were randomized 1:1 to active or sham stimulation. After the 5th post-implant month, all subjects received responsive stimulation in an open label period (OLP) to complete 2 years of post-implant follow-up. All 191 subjects were randomized. The percent change in seizures at the end of the blinded period was -37.9 % in the active and -17.3 % in the sham stimulation group ($p = 0.012$, Generalized Estimating Equations). The median percent reduction in seizures in the OLP was 44 % at 1 year and 53 % at 2 years, which represents a progressive and significant improvement with time ($p < 0.0001$). The investigators reported that serious adverse event rate was not different between subjects receiving active and sham stimulation. Adverse events were consistent with the known risks of an implanted medical device, seizures, and of other epilepsy treatments. There were no adverse effects on neuropsychological function or mood.

Bergey et al (2014) assessed the long-term efficacy and safety of responsive direct cortical stimulation in adults with medically refractory partial onset seizures. Adults with medically refractory partial onset seizures were treated with a cranially implanted responsive neurostimulator that delivers stimulation to 1 or 2 seizure foci via chronically implanted electrodes when specific electrocorticographic patterns are detected (RNS® System). Subjects had completed a 2-year primarily open label safety study (n = 65) or a 2-year randomized blinded controlled safety and efficacy study (n = 191); 230 subjects transitioned into an ongoing 7-year long-term study to assess safety and efficacy. The average subject was 34 years old (18 to 66) with epilepsy for 19.6 years (2 to 57). The median pre-implant frequency of disabling partial or generalized tonic clonic seizures was 10.2 seizures a month. Prior treatments included the vagus nerve stimulator (32 %) and epilepsy surgery (34 %). Mean post-implant follow-up was 4.7 years (5 weeks to 8.6 years) with an accumulated experience of 1,199 patient implant years and 1,107 patient stimulation years. The median percent seizure reduction in the randomized blinded controlled trial at 1 year was 44 % and at 2 years was 53 % ($p < 0.0001$ GEE) and ranged from 55 % to 60 % over post-implant years 3 through 6 for patients followed in the long-term study. Significant improvements in quality of life (QOL) were maintained (p <

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0.05). The most common serious adverse events related to the device in all studies combined were implant site infection (8.2 %) and neurostimulator explantation (3.9 %).

Patients with RNS Stimulators cannot undergo magnetic resonance imaging (MRI) procedures, nor can they undergo diathermy procedures, electro-convulsive therapy (ECT) or transcranial magnetic stimulation (TMS). The energy created from these procedures can be sent through the neurostimulator and cause permanent brain damage, even if the device is turned off. The most frequent adverse events reported in clinical trials of the Neuropace were implant site infection and premature battery depletion.

The AAN's practice parameter on "Temporal lobe and localized neocortical resections for epilepsy" (Engel et al, 2003) supported surgery (including amygdalohippocampectomy) for refractory TLE.

Maguire et al (2011) stated that "There is consensus that amygdalohippocampectomy is likely to be beneficial for people with drug-resistant temporal lobe epilepsy".

Kuang et al (2014) noted that TLE is a recurrent chronic nervous system disease. The conventional treatment is medicine. So far, anterior temporal lobectomy (ATL) and selective amygdalohippocampectomy (SAH; removal of the amygdala and hippocampus only) are becoming the 2 main approaches. These investigators compared the therapeutic effects between SAH and ATL in the treatment of TLE. They conducted a meta-analysis of published RCTs. The review applied the search strategy developed by the Cochrane Epilepsy Group and the Rev. Man 5.0 software to analyze. These researchers also drew the forest plots with Risk Ratio (RR) as effect size. A total of 6 studies were eligible, with a total of 626 patients (337 patients with SAH and 289 patients with ATL). There was no statistical significance of post-operative seizure control rate after 1 year, as well as the increase rate and decrease rate of verbal memory function between SAH and ATL. There is no statistical difference of therapeutic effects between SAH and ATL in the treatment of TLE. The authors concluded that it is advised that clinically, physicians should choose the appropriate approach according to operation indications to improve the results of post-operative recovery.

Kovanda et al (2014) stated that a number of different surgical techniques are effective for treatment of drug-resistant MTLE. Of these, trans-sylvian SAH, which was originally developed to maximize temporal lobe preservation, is arguably the most technically demanding to perform. Recent studies have suggested that SAH may result in better neuropsychological outcomes with similar post-operative seizure control as standard ATL, which involves removal of the lateral temporal neocortex. These investigators described technical nuances to improve the safety of SAH. Wide sylvian fissure opening and use of neuro-navigation allows an adequate exposure of the amygdala and hippocampus through a corticotomy within the inferior insular sulcus. Avoidance of rigid retractors and careful manipulation and mobilization of middle cerebral vessels will minimize ischemic complications. Identification of important landmarks during amygdalohippocampectomy, such as the medial edge of the tentorium and the third nerve within the intact arachnoid membranes covering the brainstem, further avoids operator disorientation. The authors concluded that SAH is a safe technique for resection of medial temporal lobe epileptogenic foci leading to drug-resistant MTLE.

Malikova et al (2014) compared 2 different surgical approaches, standard microsurgical ATL and stereotactic radiofrequency SAHE for MTLE, with respect to the extent of resection or destruction, clinical outcomes, and complications. A total of 75 MTLE patients were included: 41 treated by SAH (11 right-sided, 30 left-sided) and 34 treated by ATL (21 right-sided, 13 left-sided). SAH and ATL seizure control were comparable (Engel I in 75.6 and 76.5 % 2 years after surgery and 79.3 and 76.5 % 5 years after procedures, respectively). The neuropsychological results of SAH patients were better than in ATL. In SAH patients, no memory deficit was found. Hippocampal (60.6 ± 18.7 %) and amygdalar (50.3 ± 21.9 %) volume reduction by SAH was significantly lower than by ATL (86.0 ± 12.7 % and 80.2 ± 20.9 %, respectively). The overall rate of surgical non-silent complications without permanent neurological deficit after ATL was 11.8 %, and another 8.8 % silent infarctions were found on MRI. The rate of clinically manifest complications after SAH was 4.9 %. The rate of visual field defects after SAH was expectably less frequent than after ATL. The authors concluded that seizure control by SAH was comparable to ATL. However, SAH was safer with better neuropsychological results.

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Jobst and Cascino (2015) reviewed resective surgery outcomes for focal epilepsy to identify which patients benefit the most. These investigators noted that similar procedures such as selective amygdalohippocampectomy and temporal lobectomy for TLE were associated with subtle differences in seizure and neuropsychological outcome.

Laser Amygdalohippocampectomy

Willie and colleagues (2014) described technical and clinical outcomes of stereotactic laser amygdalohippocampotomy with real-time MR thermal imaging guidance. With patients under general anesthesia and using standard stereotactic methods, a total of 13 adult patients with intractable MTLE (with and without mesial temporal sclerosis [MTS]) prospectively underwent insertion of a saline-cooled fiber-optic laser applicator in amygdalohippocampal structures from an occipital trajectory. Computer-controlled laser ablation was performed during continuous MR thermal imaging followed by confirmatory contrast-enhanced anatomic imaging and volumetric reconstruction. Clinical outcomes were determined from seizure diaries. A mean 60 % volume of the amygdalohippocampal complex was ablated in 13 patients (9 with MTS) undergoing 15 procedures. Median hospitalization was 1 day. With follow-up ranging from 5 to 26 months (median of 14 months), 77 % (10/13) of patients achieved meaningful seizure reduction, of whom 54 % (7/13) were free of disabling seizures. Of patients with pre-operative MTS, 67 % (6/9) achieved seizure freedom. All recurrences were observed before 6 months. Variances in ablation volume and length did not account for individual clinical outcomes. Although no complications of laser therapy itself were observed, 1 significant complication, a visual field defect, resulted from deviated insertion of a stereotactic aligning rod, which was corrected before ablation. The authors concluded that real-time MR-guided stereotactic laser amygdalohippocampotomy is a technically novel, safe, and effective alternative to open surgery. They stated that further evaluation with larger cohorts over time is needed.

Mathon and associates (2015) reviewed the published literature related to the outcome of the surgical treatment of MTLE associated with hippocampal sclerosis (HS) and described the future prospects in this field. Surgery of MTLE associated with HS achieves long-term seizure freedom in about 70 % (62 to 83 %) of cases. Seizure outcome is similar in the pediatric population. Mortality following temporal resection is very rare (less than 1 %) and the rate of definitive neurological complication is low (1 %). Gamma knife stereotactic radiosurgery used as a treatment for MTLE would have a slightly worse outcome to that of surgical resection, but would provide neuropsychological advantage. However, the average latency before reducing or stopping seizures is at least 9 months with radiosurgery. Regarding palliative surgery, amygdalohippocampal stimulation has been demonstrated to improve the control of epilepsy in carefully selected patients with intractable MTLE who are not candidates for resective surgery. Recent progress in the field of imaging and image-guidance should allow to elaborate tailored surgical strategies for each patient in order to achieve seizure freedom. Concerning therapeutics, closed-loop stimulation strategies allow early seizure detection and responsive stimulation. It may be less toxic and more effective than intermittent and continuous neuro-stimulation. Moreover, stereotactic radiofrequency amygdalohippocampectomy is a recent approach leading to hopeful results. Closed-loop stimulation and stereotactic radiofrequency amygdalohippocampectomy may provide a new treatment option for patients with drug-resistant MTLE. The authors concluded that mesial temporal lobe surgery has been widely evaluated and has become the standard treatment for MTLE associated with HS. Alternative surgical procedures like gamma knife stereotactic radiosurgery and amygdalohippocampal stimulation are currently under assessment, with promising results.

Chang et al (2015) noted that surgery can be a highly effective treatment for medically refractory TLE. The emergence of minimally invasive resective and non-resective therapeutic options has led to interest in epilepsy surgery among patients and providers. Nevertheless, not all procedures are appropriate for all patients, and it is critical to consider seizure outcomes with each of these approaches, as seizure freedom is the greatest predictor of patient quality of life. Standard ATL remains the gold standard in the treatment of TLE, with seizure freedom resulting in 60 to 80 % of patients. It is currently the only resective epilepsy surgery supported by RCTs and offers the best protection against lateral temporal seizure onset. Selective amygdalohippocampectomy techniques preserve the lateral cortex and temporal stem to varying degrees and can result in favorable rates of seizure freedom but the risk of recurrent seizures appears slightly greater than with ATL, and it is unclear if neuropsychological outcomes are improved with selective approaches.

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Stereotactic radiosurgery presents an opportunity to avoid surgery altogether, with seizure outcomes now under investigation. Stereotactic laser thermo-ablation allows destruction of the mesial temporal structures with low complication rates and minimal recovery time, and outcomes are also under study. Finally, while neuromodulatory devices such as responsive neuro-stimulation, vagal nerve stimulation, and deep brain stimulation have a role in the treatment of certain patients, these remain palliative procedures for those who are not candidates for resection or ablation, as complete seizure freedom rates are low. The authors concluded that further development and investigation of both established and novel strategies for the surgical treatment of TLE will be critical moving forward, given the significant burden of this disease.

Magnetic Resonance-Guided Laser Interstitial Thermal Therapies

Lewis and colleagues (2015) reported the feasibility, safety, and clinical outcomes of an exploratory study of magnetic resonance-guided laser interstitial thermal therapy (MRgLITT) as a minimally invasive surgical procedure for the ablation of epileptogenic foci in children with drug-resistant, lesional epilepsy. These investigators performed a retrospective chart review of all MRgLITT procedures at a single tertiary care center. All procedures were performed using a Food and Drug Administration (FDA)-cleared surgical laser ablation system (Visualase Thermal Therapy System). Pre-defined clinical and surgical variables were extracted from archived medical records. A total of 17 patients underwent 19 MRgLITT procedures from May 2011 to January 2014. Mean age at seizure onset was 7.1 years (range of 0.1 to 14.8). Mean age at surgery was 15.3 years (range of 5.9 to 20.6). Surgical substrates were mixed but mainly composed of focal cortical dysplasia (n = 11); complications occurred in 4 patients. Average length of hospitalization post-surgery was 1.56 days. Mean follow-up was 16.1 months (n = 16; range of 3.5 to 35.9). Engel class I outcome was achieved in 7 patients (7/17; 41 %), Engel class II in 1 (1/17; 6 %), Engel class III in 3 (3/17; 18 %), and Engel class IV in 6 (6/17; 35 %); 3 patients (3/8; 38 %) with class I and II outcomes and 5 patients (5/9; 56 %) with class III and IV outcomes had at least 1 prior resection. Fisher's exact test was not statistically significant for the association between Engel class outcome and previous resection (p = 0.64). The authors concluded that this study provided descriptive results regarding the use of MRgLITT in a mixed population of pediatric, lesional, drug-resistant epilepsy cases. The ability to classify case-specific outcomes and reduce technical complications is anticipated as experience develops. They stated that further multi-center, prospective studies are needed to delineate optimal candidates for MRgLITT, and larger cohorts are needed to more accurately define outcome and complication rates.

Kang et al (2016) described mesial temporal lobe ablated volumes, verbal memory, and surgical outcomes in patients with medically intractable MTLE. Treated with MRI-guided stereotactic laser interstitial thermal therapy (LiTT). These researchers prospectively tracked seizure outcome in 20 patients with drug-resistant MTLE who underwent MRI-guided LiTT from December 2011 to December 2014. Surgical outcome was assessed at 6 months, 1 year, 2 years, and at the most recent visit. Volume-based analysis of ablated mesial temporal structures was conducted in 17 patients with MTS and results were compared between the seizure-free and not seizure-free groups. Following LiTT, proportions of patients who were free of seizures impairing consciousness (including those with auras only) are as follows: 8 of 15 patients (53 %, 95 % CI: 30.1 to 75.2 %) after 6 months, 4 of 11 patients (36.4 %, 95 % CI: 14.9 to 64.8 %) after 1 year, 3 of 5 patients (60 %, 95 % CI: 22.9 to 88.4 %) at 2-year follow-up. Median follow-up was 13.4 months after LiTT (range of 1.3 months to 3.2 years). Seizure outcome after LiTT suggested an all or none response; 4 patients had anterior temporal lobectomy (ATL) after LiTT; 3are seizure-free. There were no differences in total ablated volume of the amygdalohippocampus complex or individual volumes of hippocampus, amygdala, entorhinal cortex, para-hippocampal gyrus, and fusiform gyrus between seizure-free and non-seizure-free patients. Contextual verbal memory performance was preserved after LiTT, although decline in non-contextual memory task scores were noted. The authors concluded that MRI-guided stereotactic LiTT is a safe alternative to ATL in patients with medically intractable MTLE. Individualized assessment is needed to examine if the reduced odds of seizure freedom are worth the reduction in risk, discomfort, and recovery time. Moreover , they stated that larger prospective studies are needed to confirm these preliminary findings, and to define optimal ablation volume and ideal structures for ablation.

McCracken and colleagues (2016) noted that surgery is indicated for cerebral cavernous malformations (CCM) that cause medically refractory epilepsy. Real-time magnetic resonance thermography (MRT)-guided

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stereotactic laser ablation (SLA) is a minimally invasive approach to treating focal brain lesions; SLA of CCM has not previously been described. These researchers described MRT-guided SLA, a novel approach to treating CCM-related epilepsy, with respect to feasibility, safety, imaging, and seizure control in 5 consecutive patients. Patients with medically refractory epilepsy undergoing standard pre-surgical evaluation were found to have corresponding lesions fulfilling imaging characteristics of CCM and were prospectively enrolled. Each underwent stereotactic placement of a saline-cooled cannula containing an optical fiber to deliver 980-nm diode laser energy via twist drill craniostomy; MR anatomic imaging was used to evaluate targeting prior to ablation. Magnetic resonance imaging provided evaluation of targeting and near real-time feedback regarding extent of tissue thermocoagulation. Patients maintained seizure diaries, and remote imaging (6 to 21 months post-ablation) was obtained in all patients. Imaging revealed no evidence of acute hemorrhage following fiber placement within presumed CCM; MRT during treatment and immediate post-procedure imaging confirmed desired extent of ablation. These investigators identified no adverse events or neurological deficits; 4 of 5 (80 %) patients achieved freedom from disabling seizures after SLA alone (Engel class 1 outcome), with follow-up ranging 12 to 28 months. Re-imaging of all subjects (6 to 21 months) indicated lesion diminution with surrounding liquefactive necrosis, consistent with the surgical goal of extended lesionotomy. The authors concluded that minimally invasive MRT-guided SLA of epileptogenic CCM is a potentially safe and effective alternative to open resection; additional experience and longer follow-up are needed.

LaRiviere and Gross (2016) stated that epilepsy is a common, disabling illness that is refractory to medical treatment in approximately 1/3 of patients, particularly among those with MTL epilepsy. While standard open mesial temporal resection is effective, achieving seizure freedom in most patients, efforts to develop safer, minimally invasive techniques have been underway for over 50 years. Stereotactic ablative techniques, in particular, radiofrequency (RF) ablation, were first developed in the 1960s, with refinements in the 1990s with the advent of modern computed tomography and magnetic resonance-based imaging. In the past 5 years, the most recent techniques have used MRI-guided laser interstitial thermotherapy (LITT), the development of which began in the 1980s, saw refinements in MRI thermal imaging through the 1990s, and was initially used primarily for the treatment of intra-cranial and extra-cranial tumors. The authors described the original stereotactic ablation trials, followed by modern imaging-guided RF ablation series for MTL epilepsy, and reviewed the 2 currently available MRI-guided LITT systems for their role in the treatment of MTL and other medically refractory epilepsies. These investigators noted that the use of laser ablation for mesial temporal sclerosis is only in its infancy, but its superior targeting and intra-operative feedback control makes it an exciting candidate for further investigation. A prospective trial comparing MRI-guided stereotactic laser ablation with open mesial temporal lobectomy would be instrumental in demonstrating the safety and effectiveness of this promising new technique for the treatment of epilepsy. Other clinical trial approaches will be necessary to demonstrate relative safety and effectiveness with respect to standard open resection techniques. However, it must be considered that the comparison of minimally invasive techniques is not solely to standard open techniques but also to continued medical therapy, as there is a significant number of patients as well as referring physicians who consider the risk, discomfort, or inconvenience of conventional resective surgery preclusive.

Waseem and co-workers (2017) stated that there is a new focus on minimally invasive treatments for medically refractory MTLE; and MRgLITT is one such minimally invasive procedure that utilizes MRI guidance and real-time feedback to ablate an epileptogenic focus. A total of 38 patients presenting exclusively with MTLE and no other lesions (including neoplasia), who underwent MRgLITT were reviewed. These investigators evaluated a number of outcome measures, including seizure freedom, neuropsychological performance, complications, and other considerations; 18 (53 %) had an Engel class I outcome, 10 patients had repeat procedures/operations, and 12 post-procedural complications occurred. Follow-up time ranged from 6 to 38.5 months. There was a decreased length of procedure time, hospitalization time, and analgesic requirement when compared to open surgery. The authors stated that in cases of well-localized MTLE this procedure may offer similar (albeit slightly lower) rates of seizure freedom versus traditional surgery. They concluded that MRgLITT may be an alternative treatment option for high-risk surgical patients and, more importantly, could increase referrals for surgery in patients with medically refractory MTLE, however, data are limited and long-term outcomes have not been evaluated. They stated that further investigation is needed to understand the potential of this minimally invasive technique for MTLE.

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Lagman and associates (2017) stated that MRgLITT is a novel minimally invasive modality that uses heat from laser probes to destroy tissue. Advances in probe design, cooling mechanisms, and real-time MRT have increased laser utilization in neurosurgery. The authors performed a systematic analysis of 2 commercially available MRgLITT systems used in neurosurgery:

I. the Visualase thermal therapy and
II. the NeuroBlate Systems.

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Data extraction was performed in a blinded fashion. A total of 22 articles were included in the quantitative synthesis. A total of 223 patients were identified with the majority having undergone treatment with Visualase (n = 154, 69 %). Epilepsy was the most common indication for Visualase therapy (8 studies, 47 %). Brain mass was the most common indication for NeuroBlate therapy (3 studies, 60 %). There were no significant differences, except in age, wherein the NeuroBlate group was nearly twice as old as the Visualase group ($p < 0.001$). Frame, total complications, and length-of-stay (LOS) were non-significant when adjusted for age and number of patients. The authors concluded that laser neurosurgery has evolved over recent decades; clinical indications are currently being defined and will continue to emerge as laser technologies become more sophisticated.

Hoppe and co-workers (2017) noted that in common with other stereotactic procedures, stereotactic laser thermocoagulation (SLT) promises gentle destruction of pathological tissue, which might become especially relevant for epilepsy surgery in the future. Compared to standard resection, no large craniotomy is necessary, cortical damage during access to deep-seated lesions can be avoided and interventions close to eloquent brain areas become possible. These researchers described the history and rationale of laser neurosurgery as well as the 2 available SLT systems (Visualase and NeuroBlate). Both systems are coupled with MRI and MR thermometry, thereby increasing patient safety. These investigators reported the published clinical experiences with SLT in epilepsy surgery (altogether approximately 200 cases) with respect to complications, brain structural alterations, seizure outcome, neuropsychological findings and treatment costs. They stated that the rate of seizure-free patients appeared to be slightly lower than for resection surgery; however, due to the inadequate quality of studies, the neuropsychological superiority of SLT has not yet been unambiguously demonstrated.

Shukla and colleagues (2017) noted that medically intractable epilepsy is associated with increased morbidity and mortality. For those with focal epilepsy and correlated electrophysiological or radiographic features, open surgical resection can achieve high rates of seizure control, but can be associated with neurologic deficits and cognitive effects. Recent innovations have allowed for more minimally invasive methods of surgical seizure control such as MRgLITT, which achieves the goal of ablating seizure foci while preserving neuropsychological function and offering real-time feedback and monitoring of tissue ablation. These investigators summarized the utilization of MRgLITT for mesial temporal lobe epilepsy and other seizure disorders. Based on studies of laser ablation for primary glial neoplasms in adults, MRgLITT for focal epilepsy stemming from low-grade glioneuronal tumors in children is under study. The full range of applications of MRgLITT in the context of medically refractory epilepsy is still being explored. The authors concluded that MRgLITT is a safe and effective therapeutic option for the management of medically intractable epilepsy in the adult and pediatric populations. Of particular significance is the minimally invasive nature of MRgLITT, which enables the surgical management of patients who are not good candidates for, or are otherwise averse to, open resection. Compared to other minimally invasive procedures, MRgLITT is associated with improved outcomes and better side effect profile. While open surgical procedures have demonstrated slightly higher rates of seizure freedom, MRgLITT is associated with reduced hospitalization time, decreased post-operative pain, and improved neuropsychological function. Moreover, these researchers stated that it is important to note that the studies reviewed were limited by small samples sizes and the relative novelty of the procedure. Other limitations of the currently available data include the lack of availability of long-term outcomes data and a scarcity of RCTs. They stated that future studies may seek to address these gaps while also looking at questions regarding the use of the procedure for multi-focal epilepsy and the relationship between time from diagnosis and MRgLITT efficacy.

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Kang and Sperling (2018) noted that a procedure called laser interstitial thermal ablation has been utilized to treat drug resistant epilepsy. With this technique, a probe is stereotactically inserted into a target structure responsible for seizures, such as mesial temporal lobe, hypothalamic hamartoma, or a small malformation of cortical development, and the tip is then heated by application of laser energy to ablate structures adjacent to the probe tip. This procedure has the advantage of selectively targeting small lesions responsible for seizures, and is far less invasive than open surgery with shorter hospitalization, less pain, and rapid return to normal activities. Initial results in mesial temporal lobe epilepsy are promising, with perhaps 50 % of patients becoming seizure-free after the procedure. Neuropsychological deficits appear to be reduced because of the smaller volume of ablated cortex in contrast to large resections. The authors concluded that more research must be done to establish optimal targeting of structures for ablation and selection of candidates for surgery, and more patients must be studied to better establish efficacy and adverse effect rates.

Cerebellar and Deep Brain Stimulation

In a Cochrane review, Sprengers and associates (2017) evaluated the safety, efficacy, and tolerability of DBS and cortical stimulation for refractory epilepsy based on RCTs. These investigators searched the Cochrane Epilepsy Group Specialized Register on September 29, 2015, but it was not necessary to update this search, because records in the Specialized Register are included in CENTRAL. They searched the Cochrane Central Register of Controlled Trials (CENTRAL) (the Cochrane Library 2016, Issue 11, November 5, 2016), PubMed (November 5, 2016), ClinicalTrials.gov (November 5, 2016), the World Health Organization (WHO) International Clinical Trials Registry Platform ICTRP (November 5, 2016) and reference lists of retrieved articles. They also contacted device manufacturers and other researchers in the field. No language restrictions were imposed; RCTs comparing DBS or cortical stimulation versus sham stimulation, resective surgery, further treatment with anti-epileptic drugs or other neurostimulation treatments (including vagus nerve stimulation). Four review authors independently selected trials for inclusion; 2 review authors independently extracted the relevant data and assessed trial quality and overall quality of evidence. The outcomes investigated were seizure freedom, responder rate, percentage seizure frequency reduction, adverse events (AEs), neuropsychological outcome and QOL. If additional data were needed, the study investigators were contacted. Results were analyzed and reported separately for different intra-cranial targets for reasons of clinical heterogeneity. A total of 12 RCTs were identified, 11of these compared 1 to 3 months of intra-cranial neurostimulation with sham stimulation. One trial was on anterior thalamic DBS (n = 109; 109 treatment periods); 2 trials on centromedian thalamic DBS (n = 20; 40 treatment periods), but only 1 of the trials (n = 7; 14 treatment periods) reported sufficient information for inclusion in the quantitative meta-analysis; 3 trials on cerebellar stimulation (n = 22; 39 treatment periods); 3 trials on hippocampal DBS (n = 15; 21 treatment periods); 1 trial on nucleus accumbens DBS (n = 4; 8 treatment periods); and 1 trial on responsive ictal onset zone stimulation (n = 191; 191 treatment periods). In addition, 1 small RCT (n = 6) compared 6 months of hippocampal DBS versus sham stimulation. Evidence of selective reporting was present in 4 trials and the possibility of a carry-over effect complicating interpretation of the results could not be excluded in 5 cross-over trials without any or a sufficient wash-out period. Moderate-quality evidence could not demonstrate statistically or clinically significant changes in the proportion of patients who were seizure-free or experienced a 50 % or greater reduction in seizure frequency (primary outcome measures) after 1 to 3 months of anterior thalamic DBS in (multi)focal epilepsy, responsive ictal onset zone stimulation in (multi)focal epilepsy patients and hippocampal DBS in (medial) temporal lobe epilepsy. However, a statistically significant reduction in seizure frequency was found for anterior thalamic DBS (mean difference (MD), -17.4 % compared to sham stimulation; 95 % CI: -31.2 to -1.0; high-quality evidence), responsive ictal onset zone stimulation (MD -24.9 %; 95 % CI: -40.1 to -6.0; high-quality evidence) and hippocampal DBS (MD -28.1 %; 95 % CI: -34.1 to -22.2; moderate-quality evidence). Both anterior thalamic DBS and responsive ictal onset zone stimulation did not have a clinically meaningful impact on QOL after 3 months of stimulation (high-quality evidence). Electrode implantation resulted in post-operative asymptomatic intra-cranial hemorrhage in 1.6 % to 3.7 % of the patients included in the 2 largest trials and 2.0 % to 4.5 % had post-operative soft tissue infections (9.4 % to 12.7 % after 5 years); no patient reported permanent symptomatic sequelae. Anterior thalamic DBS was associated with fewer epilepsy-associated injuries (7.4 versus 25.5 %; p = 0.01) but higher rates of self-reported depression (14.8 versus 1.8 %; p = 0.02) and subjective memory impairment (13.8 versus 1.8 %; p = 0.03); there were no significant differences in formal neuropsychological testing results between the groups. Responsive ictal-onset zone stimulation appeared to

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be well-tolerated with few side effects. The limited number of patients precluded firm statements on safety and tolerability of hippocampal DBS. With regards to centromedian thalamic DBS, nucleus accumbens DBS and cerebellar stimulation, no statistically significant effects could be demonstrated but evidence is of only low to very low quality. The authors concluded that except for 1 very small RCT, only short-term RCTs on intra-cranial neurostimulation for epilepsy are available. Compared to sham stimulation, 1 to 3 months of anterior thalamic DBS ((multi)focal epilepsy), responsive ictal onset zone stimulation ((multi)focal epilepsy) and hippocampal DBS (temporal lobe epilepsy) moderately reduced seizure frequency in refractory epilepsy patients. Anterior thalamic DBS was associated with higher rates of self-reported depression and subjective memory impairment. Thee investigators stated that there is insufficient evidence to make firm conclusive statements on the safety and efficacy of hippocampal DBS, centromedian thalamic DBS, nucleus accumbens DBS and cerebellar stimulation. They stated that there is a need for more, large and well-designed RCTs to validate and optimize the safety and efficacy of invasive intra-cranial neurostimulation treatments.

High-Frequency Oscillations in Epilepsy Surgery Planning

Gloss and colleagues (2017) noted that epilepsy is a serious brain disorder characterized by recurrent unprovoked seizures. Approximately 2/3 of seizures can be controlled with anti-epileptic medications. For some of the others, surgery can completely eliminate or significantly reduce the occurrence of disabling seizures. Localization of epileptogenic areas for resective surgery is far from perfect, and new tools are being examined to more accurately localize the epileptogenic zone and improve the likelihood of freedom from post-surgical seizures. Recordings of pathological high-frequency oscillations (HFOs) may be one such tool. In a Cochrane review, these researchers evaluated the ability of HFOs to improve the outcomes of epilepsy surgery by helping to identify more accurately the epileptogenic areas of the brain. For the latest update, these investigators searched the Cochrane Epilepsy Group Specialized Register (July 25, 2016), the Cochrane Central Register of Controlled Trials (CENTRAL) via the Cochrane Register of Studies Online (CRSO, July 25, 2016), Medline (Ovid, 1946 to July 25, 2016), CINAHL Plus (EBSCOhost, July 25, 2016), Web of Science (Thomson Reuters, July 25, 2016), ClinicalTrials.gov (July 25, 2016), and the WHO International Clinical Trials Registry Platform ICTRP (July 25, 2016). They included studies that provided information on the outcomes of epilepsy surgery for at least 6 months and which used HFOs in making decisions regarding epilepsy surgery. The primary outcome of the review was the Engel Class Outcome System (class I = no disabling seizures, II = rare disabling seizures, III = worthwhile improvement, IV = no worthwhile improvement). Secondary outcomes were responder rate, ILAE epilepsy surgery outcome, frequency of AEs from any source and QOL outcomes. These researchers intended to analyze outcomes via an aggregated data fixed-effect model meta-analysis. A total of 2 studies representing 11 participants met the inclusion criteria. Both studies were small non-randomized trials, with no control group and no blinding. The quality of evidence for all outcomes was very low. The combination of these 2 studies resulted in 11 participants who prospectively used ictal HFOs for epilepsy surgery decision-making. Results of the post-surgical seizure freedom Engel class I to IV outcome were determined over a period of 12 to 38 months (average of 23.4 months) and indicated that 6 participants had an Engel class I outcome (seizure freedom), 2 had class II (rare disabling seizures), 3 had class III (worthwhile improvement); no AEs were reported. Neither study compared surgical results guided by HFOs versus surgical results guided without HFOs. The authors concluded that no reliable conclusions can be drawn regarding the efficacy of using HFOs in epilepsy surgery decision-making at present.

Feyissa and associates (2018) examined the relationship between HFOs and the presence of pre-operative seizures, WHO tumor grade, and isocitrate dehydrogenase 1 (IDH1) mutational status in gliomas. These investigators retrospectively studied intra-operative electrocorticography (ECoG) recorded in 16 patients with brain tumor (12 presenting with seizures) who underwent awake craniotomy and surgical resection between September 2016 and June 2017. The number and distribution of HFOs were determined and quantified visually and with an automated HFO detector. A total of 5 patients had low-grade (1 with grade I and 4 with grade II) and 11 had high-grade (6 with grade III and 5 with grade IV) brain tumors. An IDH1 mutation was found in 6 patients. Patients with a history of pre-operative seizures were more likely to have HFOs than those without pre-operative seizures (9 of 12 versus 0 of 4, $p = 0.02$). The rate of HFOs was higher in patients with IDH1 mutant (mean of 7.2 per minute) than IDH wild-type (mean of 2.3 per minute)

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genotype (p = 0.03). The authors concluded that HFOs were common in brain tumor-related epilepsy, and HFO rate may be a useful measure of epileptogenicity in gliomas.

The authors stated that the retrospective, single-center design of this study had inherent limitations. The small sample size (n = 16) limited definitive conclusions to be drawn regarding the association between HFOs and seizures in brain tumor-related epilepsy (BTRE). In this cohort, HFOs were not detected independently of spikes or sharp waves, although in 1 patient these researchers observed periodic sharp wave discharges without accompanying HFOs. This raised the question of whether HFO analysis added any degree of sensitivity over standard "Berger band" analysis of ECoG in this population. These observation should, however, be interpreted cautiously, particularly given the low sampling rate, which might have resulted in higher-frequency HFOs (fast ripples) being missed. Oscillations in the gamma frequency range, as seen in the majority of this cohort, have been implicated in generating ictal-like discharges in an in-vitro model of epilepsy. Moreover, locally generated gamma oscillations preceding inter-ictal discharges have been found to occur more frequently in the seizure-onset zone in non-tumoral epilepsies. Conversely, some studies suggested that fast ripples were more reliable biomarkers for the epileptogenic zone than slower-frequency oscillations. Although the distinction between the type of HFO and epileptogenicity is not absolute, it is of interest to examine if these observations endure in BTRE. Although these investigators observed that in their cohort HFO-generating tissue was completely resected (on the basis of post-operative MRI findings along with intra-operative photos of grid and/or strip placement), they did not examine the completeness of resection of HFO-generating tissue because of the lack of post-resection ECoG in the majority. However, the favorable seizure freedom outcome of the cohort (9 of 12 become seizure-free), albeit with a short follow-up period, could reflect the extent of surgery, with a majority (9 of 12) undergoing gross-total resection. Indeed, peri-tumoral tissue could be associated with subtle pathologies such as mild forms of cortical dysplasia that could be highly epileptogenic and may result in seizure recurrence if left unresected. The authors stated that future prospective studies assessing the completeness of resection of HFO-generating tissue vis-à-vis seizure freedom outcome in BTRE are needed. Moreover, they noted that given the short-term post-operative follow-up, the seizure freedom outcome of this cohort should be interpreted cautiously. Taken together, given that HFOs were seen only in those presenting with seizures, the lack of surgical tailoring using HFOs as a surrogate, and the non-controlled surgical outcome data, these findings should be interpreted cautiously; and prospective studies addressing these issues are needed to reproduce these findings and to further highlight the clinical utility of HFOs in BTRE.

Examination of Genetic Variations in Refractory Epilepsy to Guide the Selection of Surgical Candidates

Stevelink and colleagues (2018) stated that in recent years, many different DNA mutations underlying the development of refractory epilepsy have been discovered. However, genetic diagnostics are still not routinely performed during pre-surgical evaluation and reports on epilepsy surgery outcome for patients with genetic refractory epilepsy are limited. These researchers provided an overview of the literature on seizure outcome following epilepsy surgery in patients with different genetic causes of refractory epilepsy. They systematically searched PubMed and Embase prior to January 2017 and included studies describing treatment outcome following epilepsy surgery in patients with genetic causes of epilepsy. They excluded studies in which patients were described with epilepsy due to tuberous sclerosis complex or Sturge-Weber syndrome (since this extensive body of research has recently been described elsewhere) and articles in which surgery was aimed to be palliative. These researchers identified 24 eligible articles, comprising a total of 82 patients who had undergone surgery for (mainly childhood-onset) refractory epilepsy due to 15 different underlying genetic causes. The success rate of surgery varied widely across these different genetic causes. Surgery was almost never effective in patients with epilepsy due to mutations in genes involved in channel function and synaptic transmission, whereas surgery was significantly more successful regarding seizure control in patients with epilepsy due to mutations in the mTOR pathway. Patients with a lesion on MRI tended to have higher seizure freedom rates than those who were MRI-negative. The authors concluded that although the evidence is still scarce, the findings of this systematic review suggested that studying genetic variations in patients with refractory epilepsy could help guide the selection of surgical candidates.

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Furthermore, an UpToDate review on "Surgical treatment of epilepsy in adults" (Cascino, 2018) does not mention examination of genetic variants as part of surgical evaluation.

Responsive Cortical Stimulation for Focal Epilepsy

In a prospective study, Nair and colleagues (2020) examined the safety and efficacy of brain-responsive neurostimulation in adults with medically intractable focal onset seizures (FOS) over 9 years. Adults treated with brain-responsive neurostimulation in 2-year feasibility or RCTs were enrolled in a long-term prospective open-label trial (LTT) to evaluate the safety, efficacy, and QOL over an additional 7 years. Safety was assessed as AEs, efficacy as median percent change in seizure frequency and responder rate, and QOL with the Quality of Life in Epilepsy (QOLIE-89) inventory. Of 256 patients treated in the initial trials, 230 participated in the LTT. At 9 years, the median percent reduction in seizure frequency was 75 % ($p < 0.0001$, Wilcoxon signed rank), responder rate was 73 %, and 35 % had a greater than or equal to 90 % reduction in seizure frequency. These investigators found that 18.4 % (47 of 256) experienced greater than or equal to 1 year of seizure freedom, with 62 % (29 of 47) seizure-free at the last follow-up and an average seizure-free period of 3.2 years (range of 1.04 to 9.6 years). Overall QOL and epilepsy-targeted and cognitive domains of QOLIE-89 remained significantly improved ($p < 0.05$). There were no serious AEs related to stimulation, and the sudden unexplained death in epilepsy (SUDEP) rate was significantly lower than pre-defined comparators ($p < 0.05$, 1-tailed $\chi 2$). The authors concluded that adjunctive brain-responsive neurostimulation provided significant and sustained reductions in the frequency of FOS with improved QOL. Stimulation was well-tolerated; implantation-related AEs were typical of other neurostimulation devices; and SUDEP rates were low. This study provided Class IV evidence that brain-responsive neurostimulation significantly reduced focal seizures with acceptable safety over 9 years. Moreover, these researchers stated that future research will examine methods by which brain-responsive neurostimulation can be optimized for individual patients with medically intractable epilepsy. With machine and deep learning techniques, clinical and electrocorticographic data features may be identified that can direct personalized neurostimulator detection and stimulation programming. They stated that additional work to define the short- and long-term mechanism(s) of action may help to determine the optimal application of these devices.

Razavi and associates (2020) stated that the RNS System is a direct brain-responsive neurostimulation system that is FDA-approved for adults with medically intractable focal onset seizures based on safety and effectiveness data from controlled clinical trials. These researchers retrospectively examined the real-world safety and effectiveness of the RNS System. A total of 8 comprehensive epilepsy centers carried out a chart review of patients treated with the RNS System for at least 1 year, in accordance with the indication for use. Data included device-related serious AEs and the median percent change in disabling seizure frequency from baseline at years 1, 2, and 3 of treatment and at the most recent follow-up. A total of 150 patients met the criteria for analysis. The median reduction in seizures was 67 % (inter-quartile range [IQR] = 33 % to 93 %, n = 149) at 1 year, 75 % (IQR = 50 % to 94 %, n = 93) at 2 years, 82 % (IQR = 50 % to 96 %, n = 38) at greater than or equal to 3 years, and 74 % (IQR = 50 % to 96 %, n = 150) at last follow-up (mean = 2.3 years); 35 % of patients had a greater than or equal to 90 % seizure frequency reduction, and 18 % of patients reported being clinically seizure-free at last follow-up. Seizure frequency reductions were similar regardless of patient age, age at epilepsy onset, duration of epilepsy, seizure onset in mesial temporal or neocortical foci, MRI findings, prior intra-cranial monitoring, prior epilepsy surgery, or prior VNS treatment. The infection rate per procedure was 2.9 % (6/150 patients); 5 of the 6 patients had an implant site infection, and 1 had osteomyelitis. Lead revisions were needed in 2.7 % (4/150), and 2.0 % (3/150) of patients had a subdural hemorrhage, none of which had long-lasting neurological consequences. The authors concluded that in this real-world experience, safety was similar and clinical seizure outcomes exceeded those of the prospective clinical trials, corroborating effectiveness of this therapy and suggesting that clinical experience has informed more effective programming.

Gellar et al (2017) examined the seizure-reduction response and safety of mesial temporal lobe (MTL) brain-responsive stimulation in adults with medically intractable partial-onset seizures of mesial temporal lobe origin. Subjects with mesial temporal lobe epilepsy (MTLE) were identified from prospective clinical trials of a brain-responsive neurostimulator (RNS System, NeuroPace). The seizure reduction over years 2 to 6 post-implantation was calculated by evaluating the seizure frequency compared to a pre-implantation

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baseline. Safety was evaluated based on reported adverse events (AEs). There were 111 subjects with MTLE; 72 % of subjects had bilateral MTL onsets and 28 % had unilateral onsets. Subjects had 1 to 4 leads placed; only 2 leads could be connected to the device. A total of 76 subjects had depth leads only, 29 had both depth and strip leads, and 6 had only strip leads. The mean follow-up was 6.1 ± (standard deviation) 2.2 years. The median percent seizure reduction was 70 % (last observation carried forward); 29 % of subjects experienced at least 1 seizure-free period of 6 months or longer, and 15 % experienced at least 1 seizure-free period of 1 year or longer. There was no difference in seizure reduction in subjects with and without mesial temporal sclerosis (MTS), bilateral MTL onsets, prior resection, prior intra-cranial monitoring, and prior vagus nerve stimulation. Furthermore, seizure reduction was not dependent on the location of depth leads relative to the hippocampus. The most frequent serious device-related AE was soft tissue implant-site infection (overall rate, including events categorized as device-related, uncertain, or not device-related: 0.03 per implant year, which was not greater than with other neurostimulation devices). The authors concluded that brain-responsive stimulation represented a safe and effective therapeutic option for patients with medically intractable epilepsy, including patients with unilateral or bilateral MTLE who were not candidates for temporal lobectomy or who have failed a prior MTL resection. Moreover, these researchers stated that although responsive stimulation should be considered palliative, many subjects experience prolonged periods without seizures. They stated that future studies and additional experience with a variety of stimulation approaches should refine and further improve the response to treatment. The current clinical experience indicated that responsive stimulation offered a much-needed therapeutic option for patients with medically intractable MTLE.

The authors stated that this study had several drawbacks. First, the trials were not powered to provide assessments in subsets of subjects; therefore, more data are needed to confirm the comparisons of seizure reduction by clinical demographic characteristic. Second, the identification of subjects as having MTS was based on physician report and not a standardized imaging protocol. Third, because subjects with MTS could have negative MRI findings and positive histopathology, some subjects with MTS may not have been identified. Fourth, the data were collected as part of an open-label study; therefore, the effect of anti-epileptic treatments could not be clearly defined. However, analysis of seizure reduction in the trials indicated that there was no difference in seizure response between subjects whose anti-epileptic drugs (AEDs) remained stable and subjects who had AED modification.

Nunna et al (2021) stated that responsive neuromodulation (RNS) is a therapeutic option for patients with medically refractory bilateral MTL epilepsy (MTLE). A paucity of data exists on the feasibility and clinical outcome of hippocampal-sparing bilateral RNS depth lead placements within the para-hippocampal white matter or temporal stem. In a retrospective study, these researchers examined seizure reduction outcomes with at least a 1-year follow-up in individuals with bilateral MTLE undergoing hippocampus-sparing implantation of RNS depth leads. They carried out a retrospective analysis of prospectively collected data on patients at their institution with bilateral MTLE who were implanted with RNS depth leads along the longitudinal extent of bi-temporal para-hippocampal white matter or temporal stem. Baseline and post-operative seizure frequency, previous surgical interventions, and post-implantation electrocorticography and stimulation data were analyzed. A total of 10 patients were included in the study (7 male, 3 female). Overall seizure frequency declined by a median 44.25 % at 3.13 years (standard deviation [SD] = 3.31) post-implantation; 4 patients (40 %) achieved 50 % responder rate at latest follow-up; 2 of the 4 patients with focal onset bilateral tonic-clonic seizures became completely seizure-free; 40 % of patients were previously implanted with a vagus nerve stimulator, and 20 % underwent a prior temporal lobectomy. All depth lead placements were confirmed as radiographically located in the para-hippocampal white matter or temporal stem without hippocampus violation. There were no cases of lead malposition. The authors concluded that extra-hippocampal or temporal stem white matter targeting during RNS surgery for bitemporal MTLE was feasible and allowed for electrographic seizure detection. Moreover, these researchers stated that larger controlled studies with longer follow-up are needed to validate these preliminary findings.

The authors stated that this study was limited by its retrospective design and a small sample size (n = 10). Lack of neuropsychological testing data limits prevented assessment of safety of extra-hippocampal approach with relation to memory or cognitive function. Furthermore, while the only surgical complication encountered was 1 infection, a larger sample size is needed to determine the relative risk of placing bilateral extra-hippocampal depth leads.

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Foutz and Wong (2022) noted that drug-resistant epilepsy (DRE), characterized by ongoing seizures despite appropriate trials of AEDs, affects approximately 1/3 of patients with epilepsy. Brain stimulation has recently become available as an alternative therapeutic option to reduce symptomatic seizures in short- and long-term follow-up studies. Several questions remain on how to optimally develop patient-specific treatments and manage therapy over the long-term. The authors discussed the clinical use and mechanisms of action of RNS and DBS in the treatment of epilepsy and highlighted recent advances that may both improve outcomes and present new challenges.

These investigators stated that current RNS and DBS programming strategies primarily target seizure termination or suppression over the short-term. However, chronic stimulation may modify the course of disease, which is a relatively unexplored approach to neurostimulation. The progressive improvements over time observed with both DBS and RNS suggested a potential underlying change in the brain's excitability. Neurostimulation of the anterior nucleus of the thalamus (ANT) using conventional stimulation settings in animal studies has been associated with reduced mossy fiber sprouting and decreased neurodegeneration with altered gene expression and cytokines. However, disease modification effects may be optimized with different parameter settings than those typically used to terminate or suppress seizures. While current methods in neurostimulation produce significant reductions in seizure frequency, adjusting neurostimulation to promote disease modification has the potential to provide meaningful improvements in seizure freedom, but may require novel clinical trial designs. Moreover, these researchers stated that further investigation is needed to define the effects of neurostimulation in epilepsy, optimize the patient-specific use of RNS and DBS, and guide the long-term management of brain stimulation therapies. Understanding the chronic cellular and molecular effects of neurostimulation in the brain and determining more rigorously whether brain stimulation has true disease-modifying effects may help optimize current stimulation therapies and also lead to the development of novel pharmacological or surgical therapies for DRE.

Furthermore, an UpToDate review on “Evaluation and management of drug-resistant epilepsy” (Sirven, 2023) states that “Resective epilepsy surgery is the treatment of choice for medically resistant lesional partial epilepsy as this has the most likely chance of producing remission. Further antiseizure medication trials, vagus nerve stimulation, deep brain stimulation, responsive cortical stimulation, and the ketogenic diet can reduce seizure frequency and improve quality of life but are more likely to be palliative, rather than curative, treatment options … For patients in whom epilepsy surgery is not an option or whose seizures persist after surgery, we suggest treatment trials with other antiseizure medications appropriate for their epilepsy syndrome and/or vagus nerve stimulation or responsive cortical stimulation (Grade 2C). While the chance of seizure remission with these treatments is not high, reductions in seizure frequency and improved quality of life are possible in most”.

Stem Cell Therapy

Naegele et al (2010) stated that the potential applications of stem cell therapies for treating neurological disorders are enormous. Many laboratories are focusing on stem cell treatments for diseases of the central nervous system, including amyotrophic lateral sclerosis, epilepsy, Huntington's disease, multiple sclerosis, Parkinson's disease, spinal cord injury, stroke, and traumatic brain injury. Among the many stem cell types under testing for neurological treatments, the most common are fetal and adult brain stem cells, embryonic stem cells, induced pluripotent stem cells, and mesenchymal stem cells. An expanding toolbox of molecular probes is now available to allow analyses of neural stem cell fates prior to and after transplantation. Concomitantly, protocols are being developed to direct the fates of stem cell-derived neural progenitors, and also to screen stem cells for tumorigenicity and aneuploidy. The rapid progress in the field suggested that novel stem cell therapy as well as gene therapy for neurological disorders are in the pipeline.

In a systematic review, Aligholi and colleagues (2021) examined available evidence of the potential applications and benefits of stem cell transplantation (SCT) in individuals with epilepsy and also its adverse effects in humans. Medline (accessed from PubMed), Google Scholar, and Scopus from inception to August 17, 2020 were systematically reviewed for related published manuscripts. The following key words (in the title) were used: "stem cell" AND "epilepsy" OR "seizure". Articles written in English that were human studies on SCT in individuals with epilepsy were all included. These investigators identified 6 related articles. Because of their different methodologies, performing a meta-analysis was infeasible; they included

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a total of 38 adults and 81 pediatric patients; 5 studies were single-arm human studies; there were no serious adverse events (AEs) in any of the studies. The authors concluded that while SCT appeared to be a promising therapeutic option for patients with drug-resistant epilepsy, data on its application are scarce and of low quality. Currently, clinical stem cell-based interventions are not justified. Perhaps, in the future, there will be a rigorous and intensely scrutinized clinical trial protocol with informed consent that could provide enough scientific merit and could meet the required ethical standards.

One-Stage, Limited-Resection Epilepsy Surgery for Bottom-of-Sulcus Dysplasia

Macdonald-Laurs and colleagues (2021) examined if 1-stage, limited corticectomy would control seizures in patients with MRI-positive, bottom-of-sulcus dysplasia (BOSD). These investigators reviewed clinical, neuroimaging, electrocorticography (ECoG), operative, and histopathology findings in consecutively operated patients with drug-resistant focal epilepsy and MRI-positive BOSD, all of whom underwent corticectomy guided by MRI and ECoG. A total of 38 patients with a median age at surgery of 10.2 (inter-quartile range [IQR] 6.0 to 14.1) years were included. BOSDs involved eloquent cortex in 15 patients; 87 % of patients had rhythmic spiking on pre-resection ECoG. Rhythmic spiking was present in 22 of 24 patients studied with combined depth and surface electrodes, being limited to the dysplastic sulcus in 7 and involving the dysplastic sulcus and gyral crown in 15; 68 % of resections were limited to the dysplastic sulcus, leaving the gyral crown. Histopathology was focal cortical dysplasia (FCD) type IIb in 29 patients and FCDIIa in 9. Dysmorphic neurons were present in the bottom of the sulcus but not the top or the gyral crown in 17 of 22 patients; 6 (16 %) patients required re-operation for post-operative seizures and residual dysplasia; re-operation was not correlated with ECoG, neuroimaging, or histologic abnormalities in the gyral crown. At a median 6.3 (IQR 4.8 to 9.9) years of follow-up, 33 (87 %) patients were seizure-free, 31 off anti-seizure medication. The authors concluded that BOSD could be safely and effectively resected with MRI and ECoG guidance, corticectomy potentially being limited to the dysplastic sulcus, without need for intra-cranial EEG monitoring and functional mapping. Level of Evidence = IV. These preliminary findings need to be validated by well-designed, prospective, multi-center studies.

Pre-Operative Diffusion Tensor Imaging for Planning and Predicting Neurological Functional Outcome in Pediatric Epilepsy Surgery

Szmuda and associates (2016) noted that recent years brought several experimental and clinical reports applying diffusion tensor tractography imaging (DTI) of the brain in epilepsy. These researchers examined available evidence for adding the DTI sequence to the standard diagnostic MRI protocol in pediatric epilepsy. Rapid and qualitative systematic review (RAE, Rapid Evidence Assessment), aggregating relevant studies from the recent 7 years. The PubMed database was hand-searched for records containing terms "tractography AND epilepsy". Only studies referring to children were included; studies were rated using "final quality of evidence". Out of 144 screened records, relevant 101 were aggregated and reviewed. The synthesis was based on 73 studies. Case-control clinical studies were the majority of the material and comprised 43.8 % of the material. Low “confirmability” and low “applicability” referred to 18 and 17 articles (29.5 % and 27.9 %), respectively. The sufficient quality of evidence supported performing DTI in temporal lobe epilepsy, malformations of cortical development and before a neurosurgery of epilepsy. The authors concluded that qualitative RAE provided an interim estimate of the clinical relevance of quickly developing diagnostic methods. Based on the critical appraisal of current knowledge, adding the DTI sequence to the standard MRI protocol may be clinically beneficial in selected patient groups with childhood temporal lobe epilepsy or as a part of planning for an epilepsy surgery.

Lacerda and co-workers (2020) stated that surgery is a key approach for achieving seizure freedom in children with focal onset epilepsy; however, the resection can affect or be in the vicinity of the optic radiations. Multi-shell diffusion MRI and tractography can better characterize tissue structure and provide guidance to help minimize surgical related deficits. While in adults, tractography has been used to demonstrate that damage to the optic radiations led to post-operative visual field deficits, this approach has yet to be properly examined in children. These researchers demonstrated the capabilities of multi-shell diffusion MRI and tractography in characterizing micro-structural changes in children with epilepsy pre- and post-surgery affecting the occipital, parietal or temporal lobes. DTI and the spherical mean technique (SMT)

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were used to examine the microstructure of the optic radiations. In addition, tractography was used to examine if pre-surgical reconstructions of the optic radiations overlap with the resection margin as measured using anatomical post-surgical T1-weighted MRI. Increased diffusivity in patients compared to controls at baseline was observed with evidence of decreased diffusivity, anisotropy, and neurite orientation distribution in contralateral hemisphere after surgery. Pre-surgical optic radiation tractography overlapped with post-surgical resection margins in 20/43 (46 %) children, and where visual data were available before and after surgery, the presence of overlap indicated a visual field deficit. The authors concluded that this was the 1st report in a pediatric series that highlighted the relevance of tractography for future pre-surgical evaluation in children undergoing epilepsy surgery and the usefulness of multi-shell diffusion MRI to characterize brain microstructure in these patients.

These investigators stated that while surgery still remains the most effective course of action to drug-resistant epilepsy, it is well-known that the brain shifts position during surgery, and this has been addressed using tractography during neurosurgical procedures. It has also been reported that there is degeneration of the visual system over time following damage to post-chiasmal visual pathways. In some cases, the long-time interval between the pre-surgical scan and surgery may have resulted in changes in the position of the visual pathways by the time surgery was carried out, as well as in other brain regions reflecting brain plasticity. Furthermore, it is also possible, that following surgery there may have been some morphological changes over time, namely, the occupation of the resection cavity by the remaining brain tissue. The timeframe of these changes is unclear; however, if this is the case, it is possible that brain shift has contributed to an under-estimation of the segmented resection area. In the absence of proper a validation technique other than post-mortem dissections, the comparison between pre- and intra-operative tractography reconstructions as well as the combination with other MRI modalities and electrophysiological recordings should also provide further assurance regarding the use of tractography for neurosurgical purposes. A further drawback is the absence of a complete set of ophthalmology data in all the patients studied, although in the data available a clear correspondence between optic radiation involvement and visual field defect was demonstrated. Finally, SMT is able to provide an alternative to examine microstructural changes following surgery in epilepsy when compared to traditional DTI parameters. The technique is still relatively new but does provide unambiguous assessments of tissue dispersion and microscopic anisotropy both of which were found to have changed in the post-surgical optic radiations indicating a potential re-organization phenomenon. However, these researchers stated that further work is needed to fully determine the time evolution and significance of these observations.

Leon-Rojas and colleagues (2021) stated that DTI is a useful neuroimaging technique for surgical planning in adult patients; however, no systematic review has been carried out to examine its use for pre-operative analysis and planning of pediatric epilepsy surgery. In a systematic review, these investigators examined the benefit of pre-operative DTI in predicting and improving neurological functional outcome after epilepsy surgery in children with intractable epilepsy. They conducted a systematic review of articles in English using PubMed, Embase and Scopus databases, from inception to January 10, 2020. All studies that used DTI as either predictor or direct influencer of functional neurological outcome (motor, sensory, language and/or visual) in pediatric epilepsy surgical candidates were included. Data extraction was carried out by 2 blinded reviewers. Risk of bias of each study was determined using the QUADAS 2 Scoring System. A total of 13 studies were included (6 case reports/series, 5 retrospective cohorts, and 2 prospective cohorts) with a total of 229 patients; 7 studies reported motor outcome; 3 reported motor outcome prediction with a sensitivity and specificity ranging from 80 % to 85.7 %, and 69.6 % to 100 %, respectively; 4 studies reported visual outcome. In general, the use of DTI was associated with a high degree of favorable neurological outcomes following epilepsy surgery. The authors concluded that the findings of this systematic review showed that the DTI can be a useful tool to determine the course and localization of white matter tracts and help in creating a tailored plan for better functional outcome; however, due to the heterogeneity of the included patients as well as the tractography parameters, a categorical recommendation could not be given. DTI appeared to have an advantage in predicting post-surgical motor outcome; however, prospective studies with longer follow-up periods are needed to examine if motor strength remains after the initial post-surgical period or if patients with poor immediate outcomes undergo some degree of recovery. Moreover, such studies might be also useful to determine surrogate markers for quantitative DTI in order to predict other functional outcomes such as language and vision preservation. Nonetheless, this review revealed that there is an unexplained lack of focus regarding tractography techniques in pediatric epilepsy patients and that the

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available studies do not include the use of advanced acquisition techniques such as diffusion spectrum imaging, constrained spherical deconvolution, among others. These researchers stated that further studies including better techniques to ensure a low rate of false-positive fiber reconstruction are needed to obtain the full picture for its use among pediatric patients, as this is a desirable field of study because these types of techniques offer a non-invasive approach which is attractive in this type of patients.

Stereotactic Radiosurgery for the Treatment of Small-Volume Hypothalamic Hamartomas with Intractable Epilepsy

Regis et al (2017) stated that epilepsies associated with hypothalamic hamartomas (HHs) are frequently drug-resistant with severe psychiatric and cognitive co-morbidities. In a prospective study, these investigators examined the safety and effectiveness of Gamma Knife radiosurgery (GKS) in the treatment of HHs with severe epilepsies. Between October 1999 and October 2007, a total of 57 patients were examined, included and treated by GKS in Timone University Hospital. Pre-operative workup and 3-year post-operative evaluation consisted of seizure diary, neuropsychological, psychiatric, endocrinologic, visual field, and visual acuity (VA) examinations. Follow-up of longer than 3 years was available for 48 patients. Topologic type was type I in 11 patients, type II in 15, type III in 17, type IV in 1, type V in 1, type VI in 1, and mixed type in 2. The median marginal dose was 17 Gy (min of 14 and max of 25 Gy). The median target volume was 398 mm3 (28 to 1,600 mm3). Due to partial results, 28 patients (58.3 %) required a 2nd treatment. The median follow-up was 71 months (36 to 153 months). At last follow-up, the rate of Engel class I outcome was 39.6 %, Engel class II was 29.2 % (I+II 68.8 %), and Engel class III was 20 %. Global psychiatric co-morbidity was considered cured in 28 %, improved in 56 %, stable in 8 %, and continued to worsen in 8 %. No permanent neurologic side effect was reported (in particular, no memory deficit). Non-disabling transient poikilothermia was observed in 3 patients (6.2 %). A transient increase of seizure frequency was reported in 8 patients (16.6 %) with a median duration of 30 days (9 to 90 days). Microsurgery was proposed because of insufficient effectiveness of GKS in 7 patients (14.5 %) with a post-operative Engel class I-II in 28.6 %. This prospective trial showed very good long-term safety and effectiveness of GKS for 2 patients. Beyond seizure reduction, the improvement of psychiatric and cognitive co-morbidities along with better school performance and social functioning, being better socially integrated, having friends having a social life, working, participating to group activities turned out to be major benefits of GKS in this group of patients with frequently catastrophic epilepsy.

On behalf of the Hypothalamic Hamartoma Writing Group, Cohen et al (2011) noted that GKS has good long-term safety/effectiveness data; but requires a substantial time (up to 2 years) before positive effects are observed; and is not recommended for larger HHs.

McGonigal et al (2017) while there are many reports of radiosurgery for treatment of drug-resistant epilepsy, a literature review is lacking. In a systematic review, these investigators summarized current literature on the use of SRS for treatment of epilepsy. They carried out a literature search using various combinations of the search terms "radiosurgery", "stereotactic radiosurgery", "Gamma Knife", "epilepsy" and "seizure", from 1990 until October 2015. Level of evidence was assessed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. A total of 55 studies met inclusion criteria. Level II evidence (prospective studies) was available for the clinical indications of mesial temporal lobe epilepsy (MTLE) and HHs treated by GKS. For remaining indications including corpus callosotomy as palliative treatment, epilepsy related to cavernous malformation and extra-temporal epilepsy, only Level IV data was available (case report, prospective observational study, or retrospective case series). No Level I evidence was available. The authors concluded that based on Level II evidence, SRS was an effective treatment to control seizures in MTLE, possibly resulting in superior neuropsychological outcomes and QOL metrics in selected subjects compared to microsurgery. SRS exhibited a better risk-benefit ratio for small HHs compared to surgical methods. Delayed therapeutic effect resulting in ongoing seizures was associated with morbidity and mortality risk.

Bourdillon et al (2021) noted that HHs are aberrant masses, composed of abnormally distributed neurons and glia. Along endocrine and cognitive symptoms, they may cause epileptic seizures, including the specific gelastic and dacrystic seizures. Surgery is the treatment of drug-resistant hamartoma epilepsy, with

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associated positive results on endocrine, psychiatric, and cognitive symptoms. Recently, alternatives to open microsurgical treatment have been proposed. These investigators reviewed these techniques and compared their safety and effectiveness. Open resection or disconnection of the hamartoma, either via pterional, trans-callosal, or trans-ventricular approach, resulted in good epileptological control; however, its high complication rate, up to 30 %, limited its indications. The purely cisternal peduncular forms remain the only indication of open, pterional approach, while other strategies have been developed to overcome the neurological, endocrine, behavioral, or cognitive complications. Laser and radiofrequency (RF) thermocoagulation-based disconnection via robot-guided stereo-endoscopy has been proposed as an alternative to open microsurgical resection and stereotactic destruction. The objective is to allow safe and complete disconnection of a possibly complex attachment zone, via a single intra-parenchymal trajectory that allows multiple laser or RF probe trajectory inside the ventricle. The effectiveness was high, with 78 % of favorable outcome, and the overall complication rate was 8 %. It was especially effective in patients with isolated gelastic seizures and pure intra-ventricular hamartomas. Stereotactic radiosurgery (SRS) has proved as effective and safer than open microsurgery, with around 60 % of seizure control and a very low complication rate. Multiple stereotactic thermocoagulation showed very interesting results with 71 % of seizure freedom and 2 % of permanent complications. Stereotactic laser interstitial thermotherapy (LITT) appeared as effective as open microsurgery (from 76 % to 81 % of seizure freedom) but causes up to 20 % of permanent complications. This technique has however been highly improved by targeting only the epileptogenic onset zone in the hamartoma, as shown on pre-operative functional MRI, leading to an improvement of epilepsy control by 45 % (92 % of seizure freedom) with no post-operative morbidity. All these results suggested that the impact of the surgical procedure did not depend on purely technical matters (laser versus RF thermocoagulation or stereotactic versus robot-guided stereo-endoscopy) but relies on the understanding of the epileptic network, including inside the hamartoma, the aim being to plan an effective disconnection or lesion of the epileptogenic part while sparing the adjacent functional structures.

Wei et al (2022) noted that young patients with HHs often present with intractable epilepsy. Currently there are no established management guidelines for HH. The authors retrospectively reviewed their single-institution experience to delineate the role of SRS. A total of 7 patients with HHs (4 females; median age of 13.7 years, range of 2.5 to 25 years) with no prior resection underwent SRS between 1987 and 2022. The clinical history, epilepsy profile, radiographic findings, and neurological outcomes were characterized. HH topographical types were classified according to the Regis classification. Outcome measures included Engel seizure classification, HH response, and the need for additional surgical interventions. All patients had Engel class IV epilepsy. A Leksell Gamma Knife was used to deliver a median margin dose of 18 Gy (range of 16 to 20 Gy) to a median hamartoma volume of 0.37 cm3 (range of 0.20 to 0.89 cm3). Seizure reduction was confirmed in 6 patients, and 2 patients had regression of their hamartoma. Two patients underwent resection and/or laser interstitial thermal therapy after SRS. At follow-up, 1 patient was seizure free, 4 patients achieved Engel class II, 1 patient had Engel class III, and 1 patient had Engel class IV seizure outcomes. The authors concluded that SRS as the initial management option for HH was associated with a low risk of adverse effects. In this institutional series reviewing small-volume HHs treated with SRS, no adverse radiation effect was detected, and the majority of patients experienced seizure reduction. These investigators stated that SRS should be considered as the 1st-line treatment for seizure control in patients with small-volume HHs.

Savateev et al (2022) stated that HH is a dysplastic lesion fused with hypothalamus and followed by epilepsy, precocious puberty and behavioral disorders. Up to 50 % of patients become seizure-free after surgery; however, various complications occur in 25 % cases. Radiofrequency thermocoagulation, LITT and SRS are alternative therapeutic options. In a retrospective study, these researchers defined the indications for SRS in patients with HH and clarified the irradiation parameters. A total of 22 patients with HH and epilepsy underwent SRS at the Moscow Gamma-knife Center. This analysis included 19 patients with sufficient follow-up data. Median age of patients was 11.5 years (range of 1.3 to 25.8). The diameter of irradiated HHs ranged between 5.5 and 40.9 mm. In 8 (36 %) cases, the volume of hamartoma exceeded 3 cm3. Mean prescribed dose was 18 ± 2.0 Gy, mean prescribed isodose was 48 ± 4.2 %. Median follow-up period was 14.8 months (range of 3.4 to 96.1). A total of 3 (15.8 %) patients were seizure-free; 1 patient (5.3 %) improved dramatically after treatment with compete resolution of generalized seizures and experienced only rare emotional seizures (Engel IB); 11 patients (57.8 %) patients reported lower incidence of seizures. Severity and incidence of seizures were the same in 4 patients (21.1 %). The best results were achieved in

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mean target dose over 20 to 22 Gy, minimal target dose over 7 to 10 Gy, covering by the prescribed dose of at least 70 % to 80 % of hamartoma volume, as well as in patients with the prescribed dose of 12 Gy delivered to almost entire volume of tumor. No patient exhibited complications following SRS. The authors concluded that SRS was safe regarding neurological, endocrine or visual disturbances. Careful patient selection for SRS made it an effective option for HH-related epilepsy. The best candidates for SRS were children with seizures aged over 1 year, hamartoma less than 3 cm3 and area of fusion with hypothalamus less than 150 mm2.

Romanelli et al (2022) noted that HHs are developmental malformations that are associated with mild-to-severe drug-refractory epilepsy; SRS is an emerging non-invasive option for the treatment of small and medium-sized HH, providing good seizure outcomes without neurological complications. In a retrospective study, these investigators reported their experience treating HH with frameless LINAC SRS. They collected clinical and neuroradiological data of 10 subjects with HH-related epilepsy who underwent frameless image-guided SRS. All patients underwent single-fraction SRS using a mean prescribed dose of 16.27 Gy (range of 16 to 18 Gy). The median prescription isodose was 79 % (range of 65 to 81 Gy). The mean target volume was 0.64 cc (range of 0.26 to 1.16 cc). a total of 8 patients experienced complete or near complete seizure freedom (Engel class I and II); 5 patients achieved complete seizure control within 4 to 18 months following the treatment; 4 patients achieved Engel class II outcome, with stable results; 1 patient had a reduction of seizure burden superior to 50 % (Engel class III); 1 patient had no benefit at all (Engel class IV) and refused further treatments. Overall, at the last follow-up, 3 patients experienced class I, 5 class II, 1 class III and 1 class IV outcome. No neurological complications were reported. The authors concluded that frameless LINAC SRS provided good seizure and long-term neuro-psychosocial outcome, without the risks of neurological complications inherently associated with microsurgical resection.

Artificial Intelligence (AI)-Assisted Technologies (Convolutional Neural Network (CNN) Deep Learning)

Chang et al (2023) noted that radiological identification of TLE is crucial for diagnosis and treatment planning. TLE neuroimaging abnormalities are pervasive at the group level; however, they can be subtle and difficult to identify by visual inspection of individual scans, prompting applications of artificial intelligence (AI)-assisted technologies. These researchers examined the ability of a convolutional neural network (CNN) algorithm to classify TLE versus patients with Alzheimer's disease (AD) versus healthy controls (HC) using T1-weighted MRI scans. They used feature visualization techniques to identify regions the CNN employed to differentiate disease types. These researchers reported the following classification results: HC accuracy = 81.54 % (SD = 1.77 %), precision = 0.81 (SD = 0.02), recall = 0.85 (SD = 0.03), and F1-score = 0.83 (SD = 0.02); TLE accuracy = 90.45 % (SD = 1.59 %), precision = 0.86 (SD = 0.03), recall = 0.86 (SD = 0.04), and F1-score = 0.85 (SD = 0.04); and AD accuracy = 88.52 % (SD = 1.27 %), precision = 0.64 (SD = 0.05), recall = 0.53 (SD = 0.07), and F1 score = 0.58 (0.05). The high accuracy in identification of TLE was remarkable, considering that only 47 % of the cohort had deemed to be lesional based on MRI alone. Model predictions were also considerably better than random permutation classifications ($p < 0.01$) and were independent of age effects. The authors concluded that AI (CNN deep learning) could classify and distinguish TLE, underscoring its potential use for future computer-aided radiological assessments of epilepsy, especially for patients who do not exhibit easily identifiable TLE associated MRI features (e.g., hippocampal sclerosis).

The authors stated that this study had several drawbacks. First, this trial was restricted to a specific form of focal epilepsy and could not be generalized to all epilepsy syndromes. It will be important for future cohorts to include not only extra-temporal epilepsy but also examine if the classifier can distinguish TLE from generalized genetic epilepsy (GGE) syndromes. This latter point is important, since GGE usually is deemed to be associated with a normal MRI brain and hence “non-lesional” in nature. Again, the possibility of a subtle pattern not otherwise detectable by human visual examination may re-define the understanding of lesions in epilepsy. In fact, preliminary evidence suggested a pattern of aberrant structural organization in GGE. Second, while these investigators identified and addressed the role of age as a potential confounder given the disease groups, it will be important for future studies to examine the role of other important confounding factors such as exposure to specific medications, years of education, etc. Third, the approach

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proposed in this study could be extended to other applications within epilepsy as well as other disease models. For example, for challenging MRI-negative cases where the neurophysiological and semiological data suggest either GGE or a focal epilepsy with rapid bi-synchrony, this approach may aid in distinguishing the epilepsy syndrome. Naturally, lateralization of seizure foci is an important application of machine learning (ML) models and may help improve access for surgical evaluation of patients whose initial clinical and imaging data fail to provide a clearly lateralizing syndrome. Future investigation could use similar CNN approaches for a variety of outcomes is also of interest, ranging from response to anti-seizure medications, dietary treatments, neuromodulation, and post-operative seizure outcome, although this study did not propose its effectiveness at predicting these phenotypes. Extending this approach to other diagnostic challenges is also promising, including distinguishing Parkinsonian syndromes or different subtypes of vascular dementia as well as gauging the neuro-anatomical signatures of conditions such as traumatic brain injury (TBI), among others. Furthermore, future studies could employ additional neuropsychological tests to examine MCI, dementia, and early, mid, and late-stage AD. Fourth, since these researchers did not use any exhaustive hyper-parameter search, this model is not fully optimized. Although this is beyond the scope of this study, this is a limitation. These researchers stated that future studies can aim to optimize the model via a hyper-parameter search using techniques such as grid search, Bayesian, etc. to examine the ideal hyper-parameters such as learning rate, batch size, epoch length, etc. for this applied model as well as for the field of deep learning (DL) in medical imaging.

Kaestner et al (2023) stated that a new frontier in diagnostic radiology is the inclusion of machine-assisted support tools that facilitate the identification of subtle lesions often not visible to the human eye. Structural neuroimaging plays an essential role in the identification of lesions in patients with epilepsy, which often coincide with the seizure focus. In a proof-of-concept (POC) study, these researchers examined the potential for a CNN to determine lateralization of seizure onset in patients with epilepsy using T1-weighted structural MRI scans as input. Using a data-set of 359 patients with TLE from 7 surgical centers, these investigators examined if a CNN based on T1-weighted images could classify seizure laterality concordant with clinical team consensus. This CNN was compared with a randomized model (comparison with chance) and a hippocampal volume logistic regression (comparison with current clinically available measures). In addition, these researchers leveraged a CNN feature visualization technique to identify regions used to classify patients. Across 100 runs, the CNN model was concordant with clinician lateralization on average 78 % (SD = 5.1 %) of runs with the best-performing model achieving 89 % concordance. The CNN out-performed the randomized model (average concordance of 51.7 %) on 100 % of runs with an average improvement of 26.2 %, and out-performed the hippocampal volume model (average concordance of 71.7 %) on 85 % of runs with an average improvement of 6.25 %. Feature visualization maps showed that in addition to the medial temporal lobe, regions in the lateral temporal lobe, cingulate, and pre-central gyrus aided in classification. The authors concluded that these extra-temporal lobe features underscored the importance of whole-brain models to highlight areas worthy of clinician scrutiny during TLE lateralization. These researchers stated that this POC study demonstrated that a CNN applied to structural MRI data can visually aid clinician-led localization of epileptogenic zone, and identify extra-hippocampal regions that may require additional radiologic attention. Level of Evidence = II.

The authors stated that many drawbacks of this study were markers to guide future study. As sample sizes increase, result stability, and the CNN architecture complexity possible will also increase. The sample size of 359 patients from 7 surgical centers was still rather modest to fully leverage a CNN approach. Indeed, when designing this trial, these researchers had a choice between maximizing sample size by restricting themselves to a single imaging modality or attempting to maximize features via additional imaging modalities; they ultimately chose to maximize sample size. These findings confirmed this emphasis because although accuracies were encouraging, performance remained variable, though the CNN did consistently outperform the 2 alternative models. Centralizing data will result in an increased sample size, increased stability, and the easy availability of additional imaging modalities. Lastly, variability may be decreased by using a rigorous gold standard for seizure-onset lateralization: patients who became seizure-free from destruction (e.g., laser ablation) or removal (e.g., resection) of the presumed seizure focus. In this trial, lateralization was based on the consensus of multi-disciplinary clinical teams based on electrophysiology, clinical imaging, seizure semiology, neuropsychological evaluation, as well as patient history. However, for patients who did not achieve Engel I surgical outcomes, ambiguity remains regarding the seizure focus. These investigators stated

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that large sample sizes, additional features, and a cohort lateralized using surgical outcome will increase model stability.

In addition to increased stability, future models can also examine the value added by alternative CNN techniques. The authors found best performance with a 2.5D and each patient normalized to a group average space, which blinded the model to rich morphological information, which these researchers compensated for by including hippocampal volume as CNN input. They tried a 3D model but found that it under-performed the 2.5D model, presumably due to the 3D model over-fitting the training and validation sets at this sample size. However, a 3D model trained on a higher number of patients may implicitly learn about morphological features (e.g., hippocampal volumes), which are useful for the predictive task. Another area of importance is addressing harmonization across centers. This is a complex issue without an established best practice for CNN. In this study, these researchers took data from 7 surgical centers with minimal harmonization beyond standardized image preprocessing. More sophisticated methods are being examined and will likely increase model performance; thus, considerable work remains to fully leverage the promise of CNN approaches for use in epilepsy clinical practice.

These researchers envision CNN tools could be implemented as a collaborative tool in which the clinician could benefit from both model prediction and visualization. Neuro-radiologists would have access not only to model prediction and confidence in that prediction but also could use the feature visualization as a guide to interrogate the extent of the epileptogenic network. In addition to lateralizing within TLE, future models can identify seizure focus in TLE versus extra-TLE. Given these possible implementations, these investigators envision future studies bench-marking CNN not only against neuroradiology experts, but also comparing neuro-radiologist performance with and without access to CNN predictions and visualization. It will be crucial in these clinical contexts to ensure the model is thoroughly validated to ensure that inaccuracies do not cause delays in treatment. These investigators stated that the development of powerful automated tools to aid in clinician-directed clinical work-flows is an exciting avenue to continue to develop.

Chronic Pulvinar Stimulation

Wong et al (2023) stated that emerging neuro-modulatory treatments (e.g., DBS and RNS) have shown promise in reducing drug-resistant seizures. While centro-median thalamic nucleus and anterior thalamic nucleus stimulation have been effective in certain types of seizures, limited research has examined pulvinar nucleus stimulation for the treatment of DRE. These investigators carried out a systematic review and individual patient data analysis. Of 78 resultant studies, 5 with transient stimulation and chronic stimulation of the pulvinar nucleus were included. Of the 20 patients reviewed, 65 % of patients had temporal lobe seizures, while 20 % had temporo-occipital/occipital lobe seizures. Transient stimulation studies via stereo-electroencephalography (SEEG) showed pulvinar evoked potential response rates of 80 % in the mesial temporal region, 76 % in the temporal neocortex, and 67 % in the TP junction. Another study reported clinically less severe seizures in 62.5 % of patients with pulvinar stimulation. In chronic stimulation studies, 80 % of patients responded to RNS or DBS, and 2 of 4 patients experienced greater than 90 % seizure reduction. The pulvinar nucleus of the thalamus emerges as a potential target for chronic stimulation in DRE; however, knowledge regarding pulvinar connectivity and chronic stimulation remains limited. The authors concluded that data on chronic pulvinar stimulation is limited; further research is needed to demonstrate clinical significance of the pulvinar for epilepsy treatment; understanding the role of pulvinar stimulation and its cortical connectivity will advance therapeutic interventions for epilepsy patients.

Robotic Stereotactic Assistance (ROSA) Device for Stereo-Electroencephalography Electrode Implantation

Gomes et al (2023) noted that robotic assistance has improved electrode implantation precision in SEEG for patients with DRE. In a systematic review and meta-analysis, these researchers examined the relative safety of the robotic-assisted (RA) procedure compared to the traditional hand-guided one. They carried out a systematic search on PubMed, Web of Science, Embase, and Cochrane for studies directly comparing robot-assisted versus manually-guided SEEG for the treatment of refractory epilepsy. The primary outcomes included target point error (TPE), entry point error (EPE), time of implantation of each electrode, operative

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time, post-operative intra-cranial hemorrhage (ICH), infection, and neurologic deficit. These investigators included 427 patients from 11 studies, of whom 232 (54.3 %) underwent RA surgery and 196 (45.7 %) underwent manually-guided surgery. The primary endpoint, TPE, was not statistically significant (MD 0.04 mm; 95 % CI: - 0.21 to - 0.29; p = 0.76). However, EPE was significantly lower in the intervention group (MD - 0.57 mm; 95 % CI: - 1.08 to - 0.06; p = 0.03). Total operative time was significantly lower in the RA group (MD - 23.66 mins; 95 % CI: - 32.01 to - 15.31; p < 0.00001), as well as the individual time of implantation of each electrode (MD - 3.35 mins; 95 % CI: - 3.68 to - 3.03; p < 0.00001). Post-operative ICH did not differ between the 2 groups: robotic (9/145; 6.2 %) versus manual (8/139; 5.7 %) (RR 0.97; 95 % CI: 0.40 to 2.34; p = 0.94). There was no statistically relevant difference in infection (p = 0.4) and post-operative neurological deficit (p = 0.47) incidence between the 2 groups. The authors concluded that there is a potential relevance in the RA procedure when comparing the traditional one, since operative time, time of implantation of each electrode, as well as EPE were significantly lower in the robotic group. Moreover, these researchers stated that further investigation is needed to corroborate the superiority of this novel technique.

In a systematic review and meta-analysis, Kaewborisutsakul et al (2024) examined the safety and effectiveness of using the robotic stereotactic assistance (ROSA) device (Zimmer Biomet; Warsaw, IN) for SEEG electrode implantation in patients with DRE. Based on the PRISMA guidelines, a literature search was conducted. A total of 855 non-duplicate relevant articles were determined, and 15 of them were selected for analysis. The benefits of the ROSA device use in terms of electrode placement accuracy, as well as operative time length, peri-operative complications, and seizure outcomes, were evaluated. Studies that were included reported on a total of 11,257 SEEG electrode implantations. The limited number of comparative studies hindered the comprehensive evaluation of the electrode implantation accuracy. Compared with frame-based or navigation-assisted techniques, ROSA-assisted SEEG electrode implantation provided significant benefits for reduction of both overall operative time (MD, -63.45 mins; 95 % CI: -88.73 to -38.17 mins; p < 0.00001), and operative time per implanted electrode (MD, -8.79 mins; 95 % CI: -14.37 to -3.21 mins; p = 0.002). No significant differences were observed in peri-operative complications (specifically, ICH) and seizure outcomes following the use of the ROSA device and other techniques for electrode implantation. The authors concluded that the available evidence demonstrated that the ROSA device is a safe and effective surgical tool for trajectory-guided SEEG electrode implantation in patients with DRE, offering benefits for saving operative time and neither increasing the risk of peri-operative complications nor negatively impacting seizure outcomes. Moreover, these investigators stated that , given that this analysis was based on the small number of available non-comparative case-series and retrospective comparative studies of low-to-moderate quality, the findings should be carefully accepted and require additional confirmation. In addition, the authors stated that future research should be directed at the evaluation of the electrode implantation accuracy with the comparison of conventional techniques and varying robotic systems, which should be preferably realized within well-designed prospective, randomized, and multi-center studies. Level of Evidence = III.

The authors stated that this systematic review and meta-analysis had several drawbacks. First, although a literature search was carried out based on PRISMA guidelines with the use of multiple databases, search terms and their combinations were selected arbitrarily. Second, although strict inclusion and exclusion criteria allowed the selection of a highly homogeneous group of studies dedicated to the technique under investigation (ROSA-assisted SEEG electrode implantation), their overall number was too small. Third, the entire analysis was based on non-comparative case series and retrospective comparative studies of low-to-moderate quality, in which evaluation with MINORS suggested methodological flaws. Fourth, included studies occasionally showed discrepancies in reporting results, which complicated data synthesis for further statistical analysis. Fifth, a limited number of comparative studies deterred a comprehensive evaluation of the electrode implantation accuracy and reduced the statistical power of performed meta-analyses, potentially resulting in an over-estimation of the effect size. Sixth, the absence of control over confounding variables in included studies might influence the results of performed meta-analyses. Seventh, meta-regression analysis for identification of potential sources of heterogeneity among the included studies was not performed. Eighth, developments of operative techniques over time and the learning curve of the ROSA device use were not assessed.

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Translocator protein 18 kDa (TSPO) PET Imaging in the Pre-Surgical Evaluation of Patients with Drug-Resistant Focal Epilepsy

Cheval M et al (2023) noted that translocator protein 18 kDa (TSPO) PET imaging is used to monitor glial activation. Recent studies have proposed TSPO PET as a marker of the epileptogenic zone (EZ) in drug-resistant focal epilepsy (DRFE). These researchers examined the contributions of TSPO imaging using [18F]DPA-714 PET and [18F]FDG PET for localizing the EZ during pre-surgical assessment of DRFE, when phase-1 pre-surgical assessment does not provide enough information. They compared [18F]FDG and [18F]DPA-714 PET images of 23 patients who had undergone a phase-1 pre-surgical assessment, using qualitative visual analysis and quantitative analysis, at both the voxel and the regional levels. PET abnormalities (increase in binding for [18F]DPA-714 versus decrease in binding for [18F]FDG) were compared with clinical hypotheses concerning the localization of the EZ based on phase-1 pre-surgical assessment. The additional value of [18F]DPA-714 PET imaging to [18F]FDG for refining the localization of the EZ was assessed. To strengthen the visual analysis, [18F]DPA-714 PET imaging was also reviewed by 2 experienced clinicians blind to the EZ location. The study included 23 patients. Visual analysis of [18F]DPA-714 PET was significantly more accurate than [18F]FDG PET to both, showed anomalies (95.7 % versus 56.5 %, p = 0.022), and provided additional information to refine the EZ localization (65.2 % versus 17.4 %, p = 0.019). All 10 patients with normal [18F]FDG PET had anomalies when using [18F]DPA-714 PET. The additional value of [18F]DPA-714 PET appeared to be greater in patients with normal brain MRI or with neocortical EZ (especially if insula was involved). Regional analysis of [18F]DPA-714 and [18F]FDG PET provided similar results. However, using voxel-wise analysis, [18F]DPA-714 was more effective than [18F]FDG for unveiling clusters whose localization was more often consistent with the EZ hypothesis (87.0 % versus 39.1 %, p = 0.019). Non-relevant bindings were observed in 14 of 23 patients in visual analysis, and 9 patients of 23 patients in voxel-wise analysis. The authors concluded that [18F]DPA-714 PET imaging provided valuable information for pre-surgical assessments of patients with DRFE. TSPO PET could become an additional tool to aid in localizing the EZ, especially in patients with negative [18F]FDG PET. Level of Evidence = IV.

These researchers stated that this study provided preliminary evidence that [18F]DPA-714 PET imaging could provide valuable additional value to the pre-surgical assessment of patients with DRFE and could become an additional tool to aid in localizing the EZ, especially in patients with negative [18F]FDG PET, normal MRI, or neocortical epilepsy, especially insular epilepsy. However, these were only exploratory results that will require further validation. [18F]DPA-714 PET appeared to be more sensitive than [18F]FDG PET, more often consistent with electrophysiologic and seizure semiology data, and to provide additional information on the cortical areas involved in the EZ. [18F]DPA-714 PET could overcome the limitations of [18F]FDG imaging; thus, adding accuracy to the phase-1 pre-surgical assessment or even prevent invasive intra-cerebral exploration in borderline cases. These investigators stated that further comparison of [18F]DPA-714 PET with the “gold standard” for the localization of the EZ, namely., intra-cerebral EEG recording and post-operative outcomes, could enable a better understanding of the accuracy of [18F]DPA-714 PET for the EZ localization.

Intraoperative MRI in Pediatric Epilepsy Surgery for the Treatment of Focal Cortical Dysplasia

Kurzbuch et al (2025) noted that complete resection is crucial for attaining seizure freedom in children with DRE due to focal cortical dysplasia (FCD); however, identifying altered structures intra-operatively can be challenging, potentially resulting in re-operations. In a retrospective, single-center study, these researchers examined the impact of intra-operative MRI (iMRI) on pediatric epilepsy surgery for the treatment of FCD. This trial was carried out at Alder Children's Hospital from 2013 to 2022, and included children diagnosed with FCD. These investigators analyzed demographics, iMRI use, the rate of continued iMRI-guided surgery and re-operations, histology, and seizure outcomes at 1-year follow-up. A total of 32 pediatric patients (median age of 10 years, 21 males and 11 females) underwent surgery for FCD. Of them 8 (25 %) had FCD Type I, 19 (59.4 %) had FCD Type II, and 1 patient (3.1 %) had FCD Type III. iMRI was used in 27 patients (84.4 %), and 17 (63 %) of these patients required further iMRI-guided surgery. A total of 4 patients underwent re-operation in a separate session. At 1-year follow-up, seizure outcomes in the iMRI group were

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Engel I in 15 patients (71.4 %), Engel II in 4 (19 %), and Engel III and IV in 1 patient (4.8 %) each; 5 patients (15.6 %) had superficially localized lesions and underwent surgery without iMRI. None of them required re-operation, and 4 (80 %) were seizure-free at 1 year; 7 patients were lost to follow-up. The authors concluded that iMRI played an important role in identifying residual epileptogenic tissue in surgery for FCD, influencing surgical decisions and resulting in additional iMRI-guided resections. Most patients with iMRI-guided surgery attained favorable seizure outcomes (Engel I) at 1 year of follow-up. Moreover, these researchers stated that long-term follow-up is needed to validate these findings and to confirm sustained seizure control.

Radiofrequency-Thermocoagulation for the Treatment of Pediatric Epilepsy

Bottan et al (2025) carried out a systematic review on radiofrequency thermocoagulation (RF-TC) in pediatric epilepsy surgery. Furthermore, as a consequence of the low number of dedicated pediatric series, these investigators performed a pooled analysis of cases published in the literature. They conducted a literature search using PubMed and Embase that produced 432 studies. These researchers excluded studies on hypothalamic hamartomas and non-RF-TC procedures such as stereotactic radiosurgery and laser interstitial thermal ablation. Stereotactic RF-TC and SEEG-guided RF-TC procedures were included. Case series and case reports with individualized data were further reviewed and pediatric cases were extracted for pooled analysis. Patient demographics, electroclinical and neuroimaging data, procedure outcomes, responder rates as well as complications were collected. The authors' unpublished experience in pediatric SEEG-guided RF-TC was included in the pooled analysis. These researchers identified 33 studies for the literature review, 24 studies were selected for pooled analysis (93 cases); 60 patients underwent SEEG-guided RF-TC. For adult and pediatric mixed series reported a 67 % responder rate and 32 % seizure freedom at 1 year. For exclusively pediatric series, 90 % responder rate and 71 % seizure freedom. In the pooled analysis, seizure freedom was attained in 45.2 % and responder rate was 74.2 % with a mean follow-up was 25.2 months (SD ± 26.6). Complication rates were low, transient neurological deficits were reported in 18 cases (19.4 %) and no deaths were associated with RF-TC. The insula and the dominant frontal lobe were the most frequent targets for ablation. Studies included were highly heterogenous and quality of evidence was low. The authors concluded that there was little high-quality evidence on RF-TC in the pediatric population regarding its safety and effectiveness; however, available evidence suggested similar results to those observed in adult population, although more studies are needed. Most cases reviewed were extracted from heterogeneous adult and pediatric series.

Responsive Neurostimulation of the Centromedian Nucleus of the Thalamus for the Treatment of Pediatric Drug-Resistant Generalized Epilepsy

In a systematic review and meta-analysis, Rosenberg et al (2024) examined the effectiveness of RNS in pediatric patients with DRE, comparing response (50 % or greater reduction in seizure frequency) rates between patients with 2 or fewer seizure foci and those with multi-focal or generalized epilepsy. They attempted to address the gap in knowledge regarding RNS effectiveness in pediatric populations. This systematic review and meta-analysis included data from PubMed, Embase, and Web of Science through November 2023, including 17 retrospective studies and a case series of 24 patients from the authors' practice for a total of 105 aggregated patients. The inclusion criteria of patients were age of 18 or less and diagnosis of DRE. Exclusion criteria were non-human subjects and cases where RNS was not utilized to treat DRE. Study inclusion criteria were detailing the use of RNS and comparing patients with 2 or less foci with other focalities. Study exclusion criteria were failure to specify RNS lead placement or type of epilepsy. The risk of bias was assessed using the ROBINS-I tool for all non-randomized studies. Effect sizes and variances were aggregated to provide a comprehensive measure of RNS effectiveness, and heterogeneity among the studies was evaluated using I2 statistics and Cochran's Q test to assess the consistency of the findings. Statistical analyses were carried out using IBM SPSS. These researchers analyzed demographics, epilepsy history, treatment outcomes, and RNS details using descriptive and inferential statistics, including Wilcoxon-Mann-Whitney, Fisher's exact, and Chi-squared tests. A total of 17 retrospective studies and a single-center case-series study, entailing 105 pediatric patients, were analyzed. Effect sizes and CIs were calculated to quantify treatment effects. Analyses showed that RNS reduced seizure frequency across a spectrum of pediatric epilepsy syndromes, irrespective of the seizures' focal, multi-focal, or generalized origins. The

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effectiveness of RNS was not influenced by the patient's sex, age at epilepsy onset, or presence of neurological and psychiatric co-morbidities. Prior VNS surgery and the presence of an epileptic syndrome were factors associated with a lower likelihood of near-complete seizure remission with RNS, underscoring the complexities of treating patients with generalized epilepsies or previous interventional failures. The necessity of further investigations into individualized surgical strategies for patients was underscored by the mixed results of comparisons of electrode characteristics with responder rates. The authors concluded that RNS was a safe and effective treatment in pediatric patients with DRE across demographic, co-morbidity, and focality variability. Moreover, these researchers stated that prospective, randomized studies are needed to validate these findings. These investigators noted that drawbacks of this study included its reliance on retrospective studies, which introduced potential bias and limited the ability to infer causality.

Speakes et al (2025) noted that responsive neurostimulation of the centromedian nucleus of the thalamus (CM-RNS) is being studied for the treatment of drug-resistant generalized epilepsy with promising results. In a retrospective study, these investigators reported outcomes of 7 patients with pediatric-onset drug-resistant generalized epilepsy, including both GGE and Lennox-Gastaut syndrome (LGS), who underwent treatment with bilateral CM-RNS. This trial was carried out at Children's Hospital of Pittsburgh from 2020 to 2022. Improvement in seizure frequency was obtained via patient and/or care-giver reports on standardized patient questionnaires. The primary outcome measure was percent improvement in seizure frequency at time of last follow-up appointment compared to baseline seizure frequency. A total of 5 of the 7 patients (71 %) had an average 50 % or greater improvement in seizure frequency among seizure types including 4 of the 5 patients (80 %) with GGE and 1 of the 2 patients (50 %) with LGS. There were no serious AEs including post-operative infection, stroke, or device malfunction/migration. The authors concluded that the findings of this study, together with other recent studies, suggested that CM-RNS could improve seizure frequency in pediatric-onset drug-resistant generalized epilepsy; however, larger studies with longer follow-up durations as well as standardized outcome measures are needed to determine long-term effectiveness and optimal patient selection for thalamic RNS.

Epicranial Neurostimulator System

Kravalis and Schulze-Bonhage (2020) stated that non-pharmacological treatment of epilepsy is increasingly recognized as an option to improve seizure control and QOL of epilepsy patients. In particular, pharmaco-resistant patients who are not candidates for epilepsy surgery could benefit from extracranial peripheral or intracranial stimulation (deep brain stimulation of the thalami and responsive focus stimulation). These researchers reported extracranial, subgaleal patient-controlled stimulation of the epileptic focus as a procedure with low invasiveness targeting the epileptogenic brain area. The implantable device EASEE System was specially developed for subgaleal neurostimulation and modified for a patient-controlled stimulation for the purpose of this trial. It is expected to gain knowledge on the feasibility and usefulness of a closed-loop transcranial stimulation system as developed in the German Federal Ministry of Education and Research (BMBF)-funded PIMIDES project. These investigators presented the study design of PIMIDES, a trial based on patient-individualized transcranial electric neurostimulation of epileptic foci. Inclusion criteria entailed a predominant epileptic focus and pharmaco-resistance to 2 anti-epileptic drug treatments. The study was prospective, unblinded, and designed to examine the safety of subgaleal implantation and transcranial stimulation. In addition, these researchers discussed the inclusion and criteria of this trial.

Schulze-Bonhage et al (2023) stated that for the large population of individuals with drug-refractory epilepsy, alternative therapeutic approaches are needed. These investigators reported the clinical trial outcomes of a novel stimulation device, which is newly available in Europe for the treatment of patients with a predominant seizure focus. They carried out a pooled analysis of the results of 2 prospective, single-arm, multi-center studies – a pilot study to examine the feasibility of neurostimulation with the EASEE System for the treatment of medically refractory focal epilepsy (EASEE II); and a pilot study to examine the feasibility of patient-controlled neurostimulation with the EASEE System for the treatment of medically refractory focal epilepsy (PIMIDES I), examining the safety and effectiveness of epicranial focal cortex stimulation (FCS) with a novel implantable device (EASEE [Precisis]) as adjunctive treatment for adult patients with drug-refractory focal epilepsy. This study was a pooled analysis of 2 non-randomized uncontrolled trials, EASEE II and PIMIDES I, which began on January 15, 2019, and January 14, 2020, respectively, and ended on July

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28, 2021. EASEE II and PIMIDES I were the 1st in-human, prospective, single-arm trials with an 8-month evaluation period. Participants were recruited from 7 European epilepsy centers. Consecutive participants with drug-refractory focal epilepsy were enrolled. Study data were analyzed from September 29, 2021 to February 2, 2022. After a 1-month prospective baseline period, subjects were implanted with the neurostimulation device. After a 1-month post-implantation recovery period, unblinded FCS was activated using both high-frequency and direct current (DC)-like components performed via electrode arrays placed epicranially above the individual epileptic focus region. Effectiveness was evaluated by the responder rate in the 6th month of stimulation compared with baseline; safety and additional end-points were assessed after device implantation and during the stimulation period. Of the 34 adult patients enrolled at 6 German and 1 Belgian investigational site, 33 (mean [SD] age, 34.6 [13.5] years; 18 male patients [54.5 %]) received the neurostimulation device implant. A total of 32 patients underwent combined high-frequency DC-like stimulation at least until the 8-month post-implant follow-up visit. After 6 months of stimulation, 17 of 32 patients (53.1 %) were responders to treatment with at least a 50 % reduction in seizure frequency compared with baseline, corresponding to a significant median seizure reduction by 52 % (95 % CI: 37 % to 76 %; $p < 0.001$). No device- or procedure-related serious AEs were reported (0; 95 % CI: 0 % to 10.58 %). There were no significant alterations in cognition, mood, or overall QOL. The authors concluded that results of this pooled analysis of 2 non-randomized uncontrolled studies suggested that FCS with a novel neurostimulation device was associated with an effective reduction in seizure frequency in patients with drug-refractory focal epilepsy and may offer a promising therapeutic option for patients with a predominant epileptic focus.

The authors stated that drawbacks of the reported data were based on the absence of a control group; therefore, placebo effects or the influence of measured or unmeasured confounding variables could not be ruled out to have contributed to the observed reduction in seizure frequency. These researchers stated that further investigations are needed for response prediction and for a better understanding of the relative contributions of DC-like stimulation and high-frequency stimulation and the best stimulation parameters. Moreover, these investigators noted that long-term benefits and effects of FCS will be reported following the completion of ongoing trials.

Hirsch et al (2023) noted that whereas high-level evidence exists on chronic neuro-modulatory effects of different brain stimulation approaches in reducing seizure frequency, evidence for acute anti-seizure effects of electrical brain stimulation during seizures is sparse. As part of an ongoing clinical trial, these researchers implanted a patient with a novel FCS device with a Laplacian electrode placed over a pre-central focal cortical dysplasia. The baseline seizure frequency was 125 per month, consisting of focal aware sensory seizures that invariably progressed to uni- or bilateral tonic contraction and clonic jerking; and primary motor seizures. Besides an overall reduction in seizure frequency, on-demand stimulation had an immediate effect on seizures with a sensory phase, whereby 63 % to 86 % of these seizures were terminated by ictal stimulation. The authors concluded that these observations provided the first evidence that ictal self-triggered transcranial FCS could interfere with the progression of seizure semiology. These researchers stated that the development of closed-loop approaches for ictal electrical cortex stimulation with a minimally invasive device as used in this report may thus fill a therapeutic gap for patients with epilepsy.

Furthermore, an UpToDate review on “Overview of the management of epilepsy in adults” (Schachter, 2025) does not mention epicranial neurostimulation/focal cortex stimulation as a management tool.

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